

UNIVERSIDADE DE SÃO PAULO

Instituto de Ciências Matemáticas e de Computação

Asymptotics of Partition Functions of Coulomb Gases

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Dissertação de Mestrado do Programa de Pós-Graduação em
Matemática (PPG-Mat)

SERVIÇO DE PÓS-GRADUAÇÃO DO ICMC-USP

Data de Depósito:

Assinatura: _____

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Asymptotics of Partition Functions of Coulomb Gases

Master dissertation submitted to the Instituto de Ciências Matemáticas e de Computação – ICMC-USP, in partial fulfillment of the requirements of the Mathematics Graduate Program, for the degree of Master in Science. *EXAMINATION BOARD PRESENTATION COPY*

Concentration Area: Mathematics

Advisor: Prof. Dr. Guilherme Lima Ferreira da Silva

USP – São Carlos
July 2026

Ficha catalográfica elaborada pela Biblioteca Prof. Achille Bassi
e Seção Técnica de Informática, ICMC/USP,
com os dados inseridos pelo(a) autor(a)

A347a Alcantara Pimenta, João Victor
Asymptotics of Partition Functions of Coulomb
Gases / João Victor Alcantara Pimenta; orientador
Guilherme Lima Ferreira da Silva. -- São Carlos,
2026.
130 p.

Dissertação (Mestrado - Programa de Pós-Graduação
em Matemática) -- Instituto de Ciências Matemáticas
e de Computação, Universidade de São Paulo, 2026.

1. Random Matrix Theory. 2. Invariant Ensembles.
3. Orthogonal Polynomials. 4. Asymptotics. 5.
Partition Function. I. Lima Ferreira da Silva,
Guilherme, orient. II. Título.

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*"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.*

Le monde est désordre.

L'équilibre n'est pas la règle, c'est l'exception.

Et je faisais le serment de travailler pour l'ordre et l'équilibre.

Que de serments! Tu vas sourire."

G.Duhamel, Maitres, 1937

ACKNOWLEDGEMENTS

For all of those who are or were:

Your courage, patience, daring, and caring has brought me here.

To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.

For all of those who will be:

To promise that I can for you as it was done for me.

This study was financed, in part, by the São Paulo Research Foundation (FAPESP), Brasil.

Process Number # 2024/06997-1.

*“João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito.”
Te amo muito, Papai.*

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas para funções partição de gases de Coulomb**. 2026. 130 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

A Teoria de Matrizes Aleatórias (RMT) tem sido uma teoria probabilística bem sucedida no estudo de sistemas complexos. Na década de 1950, Wigner propôs que as estatísticas locais dos níveis de ressonância de espalhamento de nêutrons em núcleos pesados poderiam ser modeladas pelo comportamento estatístico dos autovalores de grandes matrizes hermitianas com invariância ortogonal, unitária ou simplética. Desde então, físicos e matemáticos perceberam que poderiam usar as estatísticas de grandes matrizes para modelar as estatísticas de sistemas anteriormente inacessíveis, como sistemas muito grandes ou complexos. Embora na física a aplicação para o espalhamento de núcleos atômicos fosse direta, os estatísticos encontraram muitas aplicações para a teoria como no estudo de matrizes de correlação, e matemáticos fizeram grandes avanços ao conectar a RMT a sistemas integráveis. Focaremos nas distribuições de autovalores cujas funções de correlação podem ser expressas como o determinante de uma função matricial de polinômios ortogonais. A função partição é uma soma sobre os microestados deste sistema de partículas e, como tal, muitas das macroquantidades, como entropia e energia, são funções da mesma. Além disso, pode ser expressa em termos dos polinômios ortogonais e, por isso, frequentemente, as macroquantidades do sistema estão diretamente relacionadas com a assintótica dos mesmos polinômios ortogonais. Tal assintótica pode ser reduzida à técnica de Laplace se o polinômio ortogonal possuir a representação integral apropriada. Neste manuscrito, contextualizaremos e replicaremos uma importante e recente técnica para derivar a assintótica da chamada função partição de uma classe de conjuntos invariantes de matrizes aleatórias com autovalores complexos.

Palavras-chave: Teoria de Matrizes Aleatórias, Ensembles Invariante de Matrizes, Polinômios Ortogonais, Processos Pontuais, Assintótica.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 130 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrix Theory (RMT) has been a successful probabilistic theory in the study of complex systems. In the 1950s, Wigner proposed that local the statistics of scattering resonance levels for neutrons off heavy nuclei could be modeled by the statistical behavior of eigenvalues of large hermitian matrices with orthogonal, unitary or symplectic invariance. Thenceforth physicist and mathematicians realized they could make use of the statistics of large matrices to model statistics of previously inaccessible systems, e.g. very large or complicated. Although in physics the application for atomic nuclei scattering was straightforward, statisticians found many uses of the theory in the study of correlation matrices and mathematicians made large advances linking RMT to integrable systems. We focus on eigenvalue distributions which correlation functions can be expressed as the determinant of a matricial function of orthogonal polynomials. The partition function is a summation over the micro-states of this particle system and as such, many of the macro-quantities, i.e. as entropy an energy, are a function of it. Moreover it can be expressed in terms of the orthogonal polynomials and therefore, oftentimes, macro-quantities of the system are directly related the asymptotic of orthogonal polynomials. Such asymptotics can be reduced to the Laplace technique if the orthogonal polynomial has a appropriate integral representation. In this manuscript we contextualize and replicate a important recent technique for deriving the asymptotics for the so-called partition function of a class of invariant ensembles of random matrices with complex eigenvalues.

Keywords: Random Matrix Theory, Invariant Matrix Ensembles, Orthogonal Polynomials, Point Processes, Asymptotics.

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INTRODUCTION

RANDOM MATRIX THEORY

EQUILIBRIUM PROBLEMS

We have seen in [Chapter 2](#) that the j.p.d.f. of eigenvalues of certain random matrices defines a determinantal point processes. We will see that not only that, but eigenvalues can have the same correlation functions as another point process: The so called *Coulomb gas*. These can be thought of as a measure on a physical gas of repelling ions restricted to some conductor under a confining potential. With this interpretation one wonders what happens to these gases in the *thermodynamic limit*. Physically, this is the regime we are used to work with; in thermodynamics we often study macroscopic properties of gases such as *entropy* or *temperature*. When considering a large enough number of particles, fluctuations of these quantities become negligible. Moreover, under the appropriate conditions, to be discussed further in the chapter, these particles are known to converge to a limit equilibrium configuration. In our matrices, these equilibrium configurations will refer to the eigenvalue univariate density function and, as discussed in [Chapter 2](#), will be expressed by the correlation kernel.

Furthermore, in this chapter we also expand on the usual formulation of minimization of a particle system's energy problems. We begin discussing *equilibrium measures* of eigenvalues - the unique measure that minimizes our functional problem. This minimization lead us to a type of problem can be solved by considering the *Euler-Lagrange*, or *Variational Equations*. We use these results to calculate the equilibrium measures for the ensemble of random matrices we work with in [Chapter 5](#). Finally, we finish this chapter doing some informal calculations that intend to motivate the asymptotic expression we get for the partition function on [Chapter 5](#). This also gives a basis estimate to which we compare the order of approximation obtained in [Byun, Kang and Seo \(2023\)](#) and explored in [Chapter 5](#).

We based ??, on equilibrium measures, on the works of [Saff and Totik \(2024\)](#) and [Deift \(1999\)](#). The argument to derive an expression for the free energy in ?? is presented in [Livan, Novaes and Vivo \(2018\)](#).

3.1 Coulomb Gases

We start by describing a familiar distribution for physicists: The Boltzmann-Gibbs distribution. This is a probability distribution that represents the probability of a given state for a system of n particles as a function of the system state's *energy* ($\mathcal{H}$) and the *temperature* ($1/(\beta n^2)$). Physically, this is the distribution that maximizes the entropy: lower energy states have higher probability of being occupied than states with higher energy. This distribution is common on thermodynamics by this very reason. Let's introduce a system that has states partitioned accordingly to such distribution. Suppose a set of n particles at positions $x_1, x_2, \dots, x_n \in \Sigma$, a *conductor* of dimension d , or in some configuration $\chi_n = (x_1, x_2, \dots, x_n)$, embedded $\mathbb{R}^{dn}$.

Definition 3.1. If dp_n is a probability measure on $(\mathbb{R}^d)^n$, we name it **Coulomb gas** if it has the form of a Boltzmann-Gibbs probability measure, namely if

$$dp_n(\chi_n) = \frac{e^{-\beta n^2 \mathcal{H}(\chi_n)}}{\mathcal{Z}_{n,\beta}^Q} \prod_{i=1}^n d^d x_i,$$

where

$$\mathcal{H}(\chi_n) = \frac{1}{2n} \sum_{i=1}^n Q(x_i) + \frac{1}{n^2} \sum_{i < j}^n g(x_i - x_j)$$

is usually called the **Hamiltonian**, or more concretely, the **configurational energy** of the system. The function $g: \Sigma \rightarrow (-\infty, \infty]$ is the *Coulomb Interacting Kernel*, solution to the Poisson equation and given by $g(x) = -\ln(|x|)$ for $d = 1, 2$. The function $Q: \mathbb{R}^d \cong \Sigma \rightarrow \mathbb{R}$ is the *external potential* and $\mathcal{Z}_{n,\beta}^Q$ is the **partition function**, or *normalization function*, given by

$$\mathcal{Z}_{n,\beta}^Q = \int_{\mathbb{R}^{dn}} e^{-\beta n^2 \mathcal{H}(\chi_n)} \prod_{i=1}^n d^d x_i.$$

The measure dp_n models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q . We wish dp_n to be a probability measure, then one must ensure that Q is such that the normalizing constant $\mathcal{Z}_{n,\beta}^Q$ is finite for all n . This is all to say that the potential should be confining: As the number of particles increases, it must counter the repulsion effect and keep limited the region where particles move around. Recall the expression for the orthogonal polynomial ensembles in ???. As we are dealing with point processes on the plane, we take the gas to be in a two dimensional conductor ($d = 2$) and $\beta = 2$. Rearranging the terms in (??) we get

$$dp_n(z_1, z_2, \dots, z_n) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)} d^2 z_i.$$

The similarity to ??? indicates that eigenvalues from invariant ensembles share some statistics with Coulomb gases. This opens the way for interpretations of the distribution of eigenvalues but also to computations using simulation methods for particle dynamics, see ???. This also motives

why most often the weight functions used for invariant ensembles will have the form, for Q polynomial,

$$w(x) = e^{-nQ(x)}.$$

This directly relates to the weight function in ?? and gives a way to trap particles in a limited set - giving smaller probabilities for configurations with more diffused particles.

Example 3.1. As explored in [Chafaï and Ferré \(2018\)](#), a few classical algorithms of simulation are very successful in replicating the measures for Coulomb gas systems. Equivalently, if one is interested in sampling eigenvalues for our invariant ensembles, one can sample from its configuration equilibrium $p_Q(\lambda)$ with respect to some potential Q using simulation of the particles. Consider for example the non-trivial case explored in [Balogh, Grava and Merzi \(2016\)](#), where, taken $d = 2$ and $n = 2$ we will have $W(x) = g(x) = \log|x|$ and impose the potential

$$Q(z) = |z|^{2a} - \operatorname{Re} t z^a.$$

This potential has disjoint intervals for some combinations of the parameters t and a - it presents a transition in $t_c \approx \sqrt{(1/a)}$. Analytically, it was just recently solved. However we can easily sample from this measure using particle simulations, as shown below. Data was taken from [Pimenta \(2024\)](#).

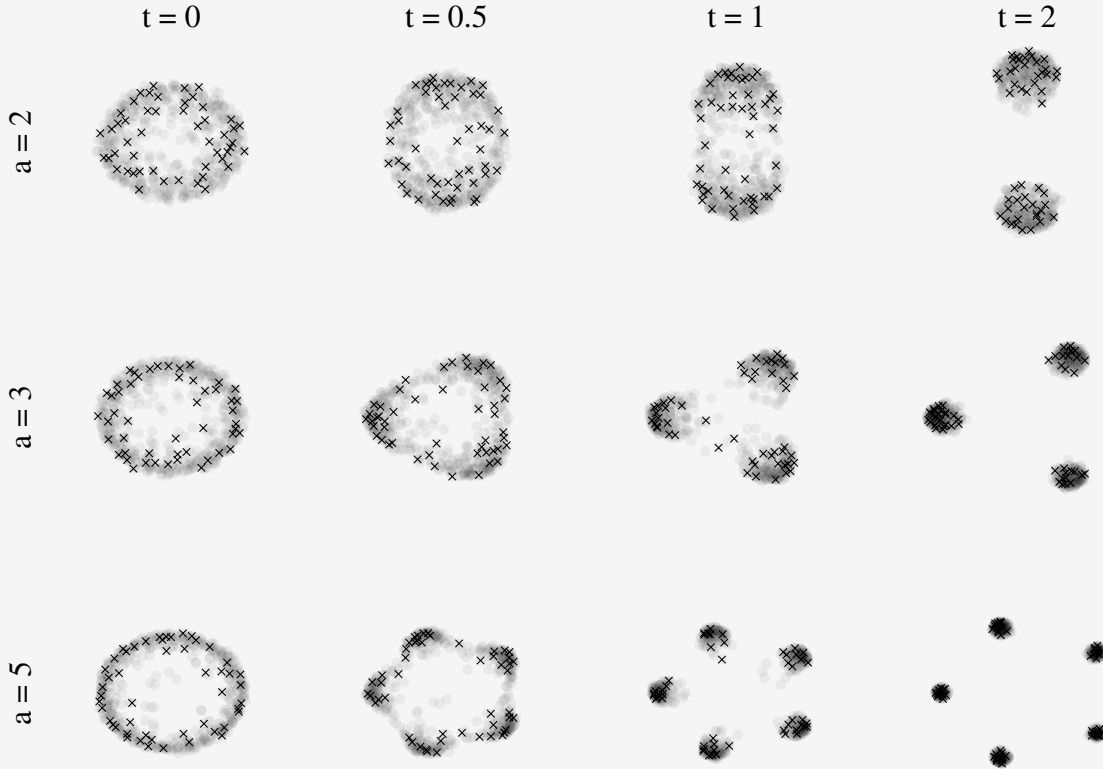


Figure 3 – Shown in the shadow is the mean configurations - in some an approximation of the limit distribution of points, and in the scatter is a random $n = 50$ sample from the distribution. We show configurations for combinations of the parameters $t = 0, 0.5, 1, 2$ from left to right and $a = 2, 3, 5$ from top to bottom.

3.2 Equilibrium Measures

The description of the eigenvalues as a *Coulomb gas* discussed in ?? and as a *point process* in ?? raises some ways to think of this system of particles. Especially, when dealing with the large n limit, the physical interpretation of this system as a interacting gas lead us to the idea of minimizing the energy of the system. In such minimization the particles should find themselves in what we call an equilibrium configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$. In the large n limit, under some mild conditions to be discussed, the counting measure defined with respect to this configuration converges to a continuous measure with compact support. This limit object is what we call an *equilibrium measure*. To clarify the point, suppose we are dealing with the j.p.d.f.

$$p_n(\chi_n) = \frac{e^{-n^2 \mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q}, \quad \mathcal{H}(\chi_n) = \frac{1}{n} \sum_{i=1}^n Q(z_i) + \frac{1}{n^2} \sum_{i \neq j}^n \ln \frac{1}{|z_i - z_j|}.$$

For any n -configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$, let μ_{χ_n} be the normalized counting measure on $\mathbb{C}$,

$$\mu_{\chi_n} = \frac{1}{n} \sum_{i=1}^n \delta_{z_i}, \quad \text{such that} \quad \int_{\mathbb{C}} d\mu_{\chi_n} = 1.$$

Then

$$\frac{1}{n^2} \sum_{i \neq j} \ln |z_i - z_j|^{-1} + \frac{1}{n} \sum_{i=1}^n Q(z_i)$$

may be written in the form

$$\int \int_{t \neq s} \ln |t - s|^{-1} d\mu_{\chi_n}(t) d\mu_{\chi_n}(s) + \int Q(s) d\mu_{\chi_n}(s).$$

This lead us, think of the convergence of $\mu_{\chi_n} \rightarrow \mu$, to the energy minimization problem

$$\begin{aligned} E^{(Q)} &= \inf_{\mu \in \mathcal{M}^1(\mathbb{C})} \left[\int \int_{t \neq s} \ln |t - s|^{-1} d\mu(t) d\mu(s) + \int Q(s) d\mu(s) \right] \\ &= \inf_{\mu \in \mathcal{M}^1(\mathbb{C})} [I_Q(\mu)] \end{aligned}$$

where

$$\mathcal{M}^1(\mathbb{C}) = \left\{ \mu \in \text{Borel measures on } \mathbb{C} \mid \int d\mu = 1 \right\}.$$

The quantity $E^{(Q)}$ is the *equilibrium energy* of the system and $I_Q(\mu_{\chi_n})$ is the *energy integral*. For μ_Q , the equilibrium measure for the minimization problem above, to be compact, Q the confining potential, must essentially be finite in such set and increase sufficiently fast around infinity. However, to analyze this case, we first consider the simplified version without an external potential. Let $\Sigma \subseteq \mathbb{C}$ be a compact subset of the complex plane and $\mathcal{M}(\Sigma)$ be the collection of all positive Borel measure of compact support in Σ . Let $\mu \in \mathcal{M}(\Sigma)$ be such a measure in the complex plane $\mathbb{C}$.

Definition 3.2. The associated **logarithmic potential** of a finite measure μ is defined by

$$U^\mu(z) := \int \ln \frac{1}{|z-t|} d\mu(t).$$

Also, $I(\mu)$, called the **logarithmic energy** of $\mu \in \mathcal{M}(\Sigma)$ is defined by

$$I(\mu) := \int \int_{z \neq t} \ln \frac{1}{|z-t|} d\mu(z) d\mu(t),$$

the **energy** V of Σ is

$$V := \inf \{I(\mu) | \mu \in \mathcal{M}(\Sigma)\}.$$

If V is finite, then the measure μ_Σ^V which minimizes the energy, is called the **equilibrium measure**.

This logarithmic potential can be easily interpreted if one reconsiders the Coulomb gas analogy discussed in ???. Note that the interaction between two particles of gas we considered is logarithmic. Then, taking the empirical counting measure μ_{χ_n} ,

$$U^{\mu_{\chi_n}}(z) = \int \ln \frac{1}{|z-t|} d\mu_{\chi_n}(t) = \sum_{i=1}^n \ln \frac{1}{|z-\lambda_i|}.$$

This is precisely the potential created by the particles interactions on the point z . There are a few useful properties one can extract about this quantity. A similar argument can also be made to justify the logarithmic energy.

Proposition 3.1. (SAFF; TOTIK, 2024) Let μ be a finite positive Borel measure of compact support on the complex plane $\mathbb{C}$. Then the logarithmic potential is super-harmonic on the whole plane, that is, it has the following properties.

1. $U^\mu(z) > -\infty$ for all z ,
2. U^μ is lower semi-continuous,
3. $U^\mu(z) \geq \frac{1}{2\pi} \int_{-\pi}^{\pi} U^\mu(z + r e^{i\theta}) d\theta$ for all z and all $r \geq 0$.

Proof. Firstly, property (1) follows directly by (2), since a lower semi-continuous function is everywhere $> -\infty$ and μ has compact support. Property (2) is proven noticing that

$$U^\mu(z) = \lim_{M \rightarrow \infty} \int \min \left(M, \ln \frac{1}{|z-t|} \right) d\mu(t).$$

The functions on the right-hand side are continuous and increasing with M , and limit of an increasing sequence of functions must be lower semi-continuous. To prove (3), note that the kernel $\ln(1/|z-t|)$ is super-harmonic in $\mathbb{C}$ for each t . If some F is analytic in a domain D and $\rho > 0$, then $|F(z)|^\rho$ is sub-harmonic in D ; furthermore, $\ln|F(z)|$ is sub-harmonic in D provided

F is not identically zero. Thus,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\theta} - t|} d\theta \leq \ln \frac{1}{|z - t|}, \quad z, t \in \mathbb{C}.$$

Using the Fubini-Toneli theorem and the inequality above

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} U^{\mu}(z + r e^{i\theta}) d\theta &= \int \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\theta} - t|} d\theta d\mu(t), \\ &\leq \int \ln \frac{1}{|z - t|} d\mu(t) = U^{\mu}(z). \end{aligned}$$

□

For technical reasons we also state ?? regarding the logarithmic energy that will become useful in a later result but to which we do not provide a proof.

Lemma 3.1. (*SAFF; TOTIK, 2024*) Let $\mu = \mu_1 - \mu_2$ be a signed Borel measure with compact support and total mass $\mu(\mathbb{C}) = 0$. Suppose further that each of the positive measures μ_1 and μ_2 has finite logarithmic energy. Then the logarithmic energy of μ

$$I(\mu) = \int \int \ln \frac{1}{|z - t|} d\mu(z) d\mu(t),$$

is non-negative and it is zero if and only if $\mu \equiv 0$.

Suppose that there is a unique measure $\mu_{\Sigma}^V \in \mathcal{M}(\Sigma)$ for which the minimum energy $V = \inf\{I(\mu) | \mu \in \mathcal{M}(\Sigma)\}$ is finite and reachable. Such a measure μ_{Σ}^V is called the *equilibrium distribution* or *equilibrium measure* of the compact set Σ . These concepts have a natural electrostatic interpretation. Suppose Σ is a conductor and charged particles are placed on it. If the force between the particles is proportional to the reciprocal of their distance and Q is some external field to which they interact to, then μ_{Σ}^V is the configuration which minimizes the energy, or maximizes the entropy, of the system. This measure gives the distribution to which this particles are in equilibrium.

We should at this point reintroduce the external potential Q . Most often than not, an external force will change the minimum energy state and equilibrium configurations. Particles tend to accumulate at the minimum of such potential. The reason why the distribution will not be a single point in space is that the logarithmic energy, from the interactions, explodes when particles are close by. For this case, a weighted version of the problem was developed. The main difference now is that Σ is not required to be compact; we only impose that there is no positive mass around infinity. There is a natural way to enforce these conditions. Let $\Sigma \subset \mathbb{C}$ be a closed set and $w : \Sigma \rightarrow [0, \infty)$. This function w is called a *weight function* on Σ . Certain weight functions make it so that there is no positive mass around infinity; we call such function *admissible*.

Definition 3.3. Define also the logarithmic capacity to be $\text{cap}(\Sigma) = e^{-V}$ with $V = \inf\{I(\mu) | \mu \in \mathcal{M}(\Sigma)\}$. A weight function $w = e^{-nQ(z)}$ is said to be **admissible** if it satisfies the following three conditions

1. Q is upper semi-continuous;
2. $\Sigma_0 := \{z \in \Sigma | Q > -\infty\}$ has positive capacity;
3. if Σ is unbounded, then $\liminf_{|z| \rightarrow \infty} \frac{Q(z)}{2 \ln(|z|)} > 1$ as $|z| \rightarrow \infty$, $z \in \Sigma$.

Let $\mathcal{M}(\Sigma)$ be the collection of all positive unit Borel measures μ with support in Σ . Our definition for the weighted energy integral now reads

$$I_Q(\mu) = \int U^\mu(s) d\mu(s) + \int Q(s) d\mu(s),$$

for $\mu \in \mathcal{M}(\Sigma)$ and is recovered if both integrals exist and are finite. Note that this is to say that

$$\begin{aligned} I_Q(\mu) &= \int U^\mu(s) d\mu(s) + \int Q(s) d\mu(s), \\ &= \int \int_{s \neq t} \ln \frac{1}{|s-t|} d\mu(s) d\mu(t) + \frac{1}{2} \left[\int Q(s) d\mu(s) + \int Q(t) d\mu(t) \right], \\ &= \int \int_{s \neq t} \left(\ln \frac{1}{|s-t|} + \frac{Q(t)}{2} + \frac{Q(s)}{2} \right) d\mu(s) d\mu(t), \\ &= \int \int_{s \neq t} \kappa^Q(s, t) d\mu(s) d\mu(t) \end{aligned}$$

converges. Denote

$$\phi^Q(x) = Q(x) - \ln(x^2 + 1)$$

and note that

$$\ln |t-s|^{-1} \geq -\frac{1}{2} \ln(1+t^2) - \frac{1}{2} \ln(1+s^2).$$

Therefore

$$\kappa^Q(s, t) \geq \frac{1}{2} \phi^Q(t) + \frac{1}{2} \phi^Q(s) \geq c_Q > -\infty$$

since $\lim_{|x| \rightarrow \infty} \phi^Q(x) = \infty$ and $\phi^Q(x) \geq c_Q > -\infty$. Thus, $I_Q \geq c_Q > -\infty$ for every $\mu \in \mathcal{M}^1(\mathbb{C})$. Remember that we are dealing with weight functions of the form $w(z) = e^{-nQ(z)}$ for $z \in \mathbb{C}$ and we want Q to be such that the weighted logarithmic energy is finite - the weight is admissible. Then, by the definition, $Q : \Sigma \rightarrow (\infty, \infty]$ must be lower semi-continuous. To avoid degeneracy, we imposed that $Q < \infty$ in some set of positive capacity. Furthermore, we must have that $\liminf_{|z| \rightarrow \infty} \frac{Q(z)}{2 \ln(|z|)} > 1$ such that the potential energy grows faster than the interaction logarithmic energy. Remember that this is similar to the condition imposed on the orthogonal polynomial ensembles in ???. On the next section we move on to the matter of determining the *equilibrium measure* μ_Q for a given external field Q .

Theorem 3.1. ([SAFF; TOTIK, 2024](#)) Let w be an admissible weight function on the closed set Σ and let

$$V_w := \inf\{I_w(\mu) | \mu \in \mathcal{M}(\Sigma)\}.$$

Then, V_w is finite and there exists a unique element $\mu_w \in \mathcal{M}(\Sigma)$ such that $I_w(\mu_w) = V_w$.

On the next section we move on to the matter of determining the *equilibrium measure* μ_Q for a given external field Q .

Example 3.2. Take the Coulomb gas measure described in ??, and restrict the conductor Σ to be the real line under the external potential

$$Q(x) = tx^{2m}.$$

By finding configurations that minimizes the Hamiltonian, one also finds the more likely configurations. These configurations, when $n \rightarrow \infty$ converges to a measure μ_Q that minimizes the logarithmic energy. It can be shown that this measure is given by

$$\text{supp } \mu_Q(x) = [-a, a], \quad \mu_Q(x) = \frac{mt}{\pi} \sqrt{a^2 - x^2} h(x),$$

where

$$a := \left(mt \prod_{l=1}^m \frac{2l-1}{2l} \right) \quad \text{and} \quad h(x) := x^{2m-2} + \sum_{j=1}^{m-1} x^{2m-2-2j} a^{2j} \prod_{l=1}^j \frac{2l-1}{2l}.$$

See [Deift \(1999\)](#) for details. Note that this simplifies to the Gaussian case if $Q(x) = x^2/2$ ($t, m = 1$). In such case we already shown that the equilibrium measure is defined by the Semicircle distribution

$$\text{supp } \mu_Q(x) = [-\sqrt{2}, \sqrt{2}], \quad \mu_Q(x) = \frac{1}{\pi} \sqrt{2 - x^2}.$$

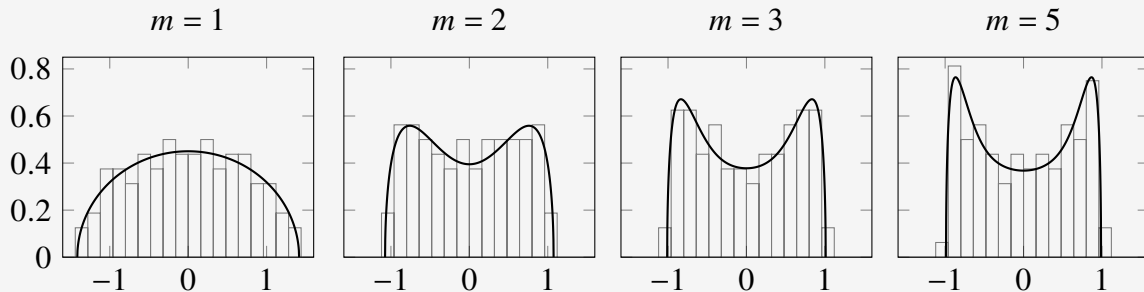


Figure 4 – Four conditions for the Monic potential, the equilibrium measure is expressed by the line plot. For all cases we used $t = 1$ and set $m = 1, 2, 3, 5$ from left to right. The bin-plot is the result of the simulation of the equivalent Coulomb Gas taken from [Pimenta \(2024\)](#).

3.2.1 Euler-Lagrange

Associated with the variational problem introduced in ?? defined by

$$\begin{aligned} E^w &= \inf_{\mu \in \mathcal{M}^1} I_Q(\mu), \\ &= \inf_{\mu \in \mathcal{M}^1} \left[\int \int \ln |t-s|^{-1} d\mu(t) d\mu(s) + \int Q(s) d\mu(s) \right], \\ &= I_Q(\mu_Q), \end{aligned}$$

one defines the so-called *Euler-Lagrange* type variational equations. They are important as they provide a tool to solve these equations for the equilibrium measure μ_Q .

Theorem 3.2. (*Variational Equations*) *The equilibrium measure μ_Q for the given variational problem in (??), where $\Sigma = \mathbb{C}$ satisfies the following conditions. There exists a real constant l such that*

1.

$$\int \left[2 \int \ln |x-y|^{-1} d\mu_Q(y) + Q(x) \right] d\tilde{\mu}(x) \geq l$$

for all $\tilde{\mu} \in \mathcal{M}^1(\mathbb{C})$ of compact support with $I_Q(\tilde{\mu}) < \infty$.

2.

$$2 \int \ln |x-y|^{-1} d\mu_Q(y) + Q(x) = l$$

for μ_Q -almost everywhere.

Conversely, if $\mu \in \mathcal{M}^1(\mathbb{C})$ has compact support, satisfies conditions (1) and (2) above, and $I_Q(\mu) < \infty$, then $\mu = \mu_Q$.

Proof. The argument of the proof is based on (DEIFT, 1999).

($\implies$) We recall that μ_Q has compact support and $I_Q(\mu_Q) = E^w < \infty$. Suppose $\mu^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I_Q(\mu^*) < \infty$. Then, defining a convex combination

$$\mu_t = t\mu^* + (1-t)\mu_Q = \mu_Q + t(\mu^* - \mu_Q) \in \mathcal{M}^1(\mathbb{C}), \quad 0 \leq t \leq 1,$$

we have

$$\begin{aligned} I_Q(\mu_t) &= \int \int \ln |t-s|^{-1} d\mu_Q(t) d\mu_Q(s) + \int Q(s) d\mu_Q(s) \\ &\quad + 2t \int \int \ln |t-s|^{-1} d\mu_Q(t) d(\mu^* - \mu_Q)(s) \\ &\quad + t \int Q(s) d(\mu^* - \mu_Q)(s) \\ &\quad + t^2 \int \int \ln |t-s|^{-1} d(\mu^* - \mu_Q)(t) d(\mu^* - \mu_Q)(s). \end{aligned}$$

Each term on the right is finite since μ_Q, μ^* have compact support and $I_Q(\mu_Q), I_Q(\mu^*) < \infty$. As $I_Q(\mu_t) \geq I_Q(\mu_Q)$ for $0 \leq t \leq 1$ by assumption of minimality and ??, it is necessary that

$$\int \left[2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) \right] d(\mu^* - \mu_Q)(x) \geq 0$$

and therefore

$$\int \left[2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) \right] d\mu^*(x) \geq l$$

where

$$\int \left[2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) \right] d\mu_Q(x) := l$$

proving the first assertion in (1). Now, let

$$B = \left\{ x \mid 2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) < l \right\}.$$

Then, B must be a bounded set of $\mathbb{C}$. Suppose $\mu^*(B) > 0$ and set

$$\mu_B^* = \frac{\chi_B}{\mu^*(B)} \mu^*$$

where χ_B is the characteristic function of the set B . Then $\mu_B^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I_Q(\mu_B^*) < \infty$. inserting $d\mu_B^*$ in (??), we find

$$l \leq \int \left[2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) \right] d\mu_B^*(x) < l$$

a contradiction. Therefore,

$$\mu^*(B) = \mu^* \left(\left\{ x \mid 2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) < l \right\} \right) = 0$$

for all measures $\mu^* \in \mathcal{M}^1(\mathbb{C})$ with support compact and $I_Q(\mu^*) < \infty$. in particular, it is true for $\mu^* = \mu_Q$. Since,

$$\begin{aligned} 0 &= \int \left[2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) - l \right] d\mu_Q(x) \\ &= \int_{\mathbb{C}-B} \left[2 \int \ln|x-y|^{-1} d\mu_Q(y) + Q(x) - l \right] d\mu_Q(x) \end{aligned}$$

and (2) follows.

($\Leftarrow$) If, now, $\mu \in \mathcal{M}^1$ satisfies (1) and (2), $I_Q(\mu) < \infty$, and suppose μ has compact support. Then write $\mu_Q = \mu + (\mu_Q - \mu)$. We have, as before

$$\begin{aligned} I_Q(\mu) &\geq I_Q(\mu_Q) \\ &= I_Q(\mu) + \int \left[2 \int \ln|x-y|^{-1} d\mu(y) + Q(x) - l \right] d(\mu_Q - \mu)(x) \\ &\quad + \int \int \ln|t-s|^{-1} d(\mu_Q - \mu)(t) d(\mu_Q - \mu)(s). \end{aligned}$$

Using (2),

$$\int \left[2 \int \ln|x-y|^{-1} d\mu(y) + Q(x) - l \right] d\mu_Q(x) - l \geq l - l = 0$$

but also the third term is strictly positive by ?? unless it is identically null. If, however, $\mu_Q - \mu \neq 0$, then we showed that

$$I_Q(\mu) \geq I_Q(\mu_Q) > I_Q(\mu),$$

a contradiction. Thus, if μ satisfies (1), (2), then it must be μ_Q . □

If, in addition we assume that

$$d\mu_Q(z) = \mu(z) d^2z$$

for some continuous function μ of compact support then the previous theorem takes the following simpler form.

Theorem 3.3. (*Strong Variational Equations*) Suppose that $d\mu_Q = \mu(z) d^2z$ for some continuous function μ of compact support. Then the conditions of ?? is replaced by

1.

$$2 \int \ln|z-t|^{-1} d\mu_Q(t) + Q(z) \geq l$$

for all $z \in \mathbb{C}$.

2.

$$2 \int \ln|z-t|^{-1} d\mu_Q(t) + Q(z) = l$$

on $\{z : \mu(z) > 0\}$.

We mentioned in (??) the possibility of knowing the density μ of the equilibrium measure μ_Q with respect to the Lebesgue measure. We will now determine μ . The support the equilibrium measure is an important quantity in determining the equilibrium measure corresponding to a weight function w . Suppose for example this support to be bounded with a smooth border and w to be a continuous function. Then, by the previous theorem, inside of the support U^{μ_w} coincides with $-Q$ plus some constant. Thus, we get a way of determining the the equilibrium logarithmic potential. From this potential, there is a way to recover the equilibrium measure. Let d^2z denote the two-dimensional area measure on $\mathbb{C}$ and Δ denote the usual Laplacian operator

$$\Delta F = \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2}.$$

Then, we have the theorem below which makes a statement about the density function.

Theorem 3.4. (*SAFF; TOTIK, 2024*) If in a region R the potential U^μ has continuous second partial derivatives, then μ is absolutely continuous in R with respect to two-

dimensional area measure d^2z , and on R we have the formula

$$d\mu = -\frac{1}{4\pi} \Delta U^\mu d^2z$$

3.2.2 Radially Symmetric Potentials

In this section we will calculate the equilibrium measure for a large class of matrices in the *random normal ensemble of matrices*. Especially, we are going to focus on ensembles such that the external potential Q is radially symmetric. This symmetry will be essential in the calculations that follows in Chapter 5. The measure calculated in this section is used to perform asymptotic analysis on the partition function for these normal ensembles. We begin by defining such ensemble.

Definition 3.4. The *random normal matrix ensemble* $(S, \mathcal{P})$ is constructed by considering the set of n -sized normal matrices with complex entries $S = \mathcal{N}_n(\mathbb{C})$ equipped with a p.d.f.

$$\mathcal{P}(\hat{M}) = e^{-n \text{Tr} Q(\hat{M})}$$

where Q is some polynomial function and $\hat{M} \in \mathcal{N}_n(\mathbb{C})$.

We first note that because we are dealing with random matrices we rewrite this p.d.f. as the following expression

$$\mathcal{P}(\hat{M}) = e^{-n \text{Tr} Q(\hat{M})} = e^{-n \sum_{i=1}^n Q(\lambda_i)}.$$

Together with ??, we say that, on the complex plane we are considering the measure on the set of eigenvalues given by

$$dp_n(z_1, \dots, z_n) = \frac{1}{\mathcal{Z}_n^Q} \prod_{j < k}^n |z_j - z_k|^2 \prod_{j=1}^n e^{-nQ(z_j)} d^2z_j.$$

We now define the potential Q to further specify the working model.

Definition 3.5. We deal with $\mathcal{E} = (S, \mathcal{P})$ a *normal ensemble* in which Q , which defines the weight function $w(x) = e^{-nQ(x)}$ is some complex potential that satisfies

1. The potential Q is radially symmetric, i.e.

$$Q(z) = q(|z|)$$

for some $q : \mathbb{R} \rightarrow \mathbb{R}$;

2. The potential Q grows sufficiently fast, i.e.

$$\liminf_{|z| \rightarrow \infty} \frac{Q(z)}{2 \ln(|z|)} > 1;$$

3. Furthermore, assume Q to be smooth, sub-harmonic in $\mathbb{C}$, and strictly sub-harmonic in a neighborhood of the support of the equilibrium measure μ_Q .

These assumptions are essential for what follows. To require radial symmetric is to say that the integral of a function of Q with respect to the area measure can be exchanged by an angle component and a line component, as all directions are homogeneous. The growth requirement comes from the need for the confinement of particles in the system - note that the interaction energy has logarithmic behavior. The conditions on sub-harmonicity can be thought of as convexity and will ensure uniqueness of the minima and connectedness of the support of the equilibrium measure. We will be dealing mainly with the real potential q as we make use of the symmetry of Q . Therefore, we also state the following proposition.

Proposition 3.2. *The real potential $q: \mathbb{R} \rightarrow \mathbb{R}$ defined as in ?? is such that $rq'(r)$ is **increasing** in the complex plane and **strictly increasing** in the support of the equilibrium measure μ_Q .*

Proof. Because Q is sub-harmonic on the plane, i.e. $\Delta Q(z) \geq 0$ for all $z \in \mathbb{C}$, then

$$\Delta Q(z) = \frac{1}{r}(rq'(r))' \geq 0 \implies rq'(r) \text{ is increasing,}$$

and strictly increasing in the support of the equilibrium measure μ_Q by the same relation with a strict inequality. $\square$

A very important feature of the partition function of normal ensembles is that it can be expressed as a summation of norming constants of the correspondent orthogonal polynomial. This was proved on ??. We have stated a few properties of the model surrounding the potential $Q(z) = q(|z|)$. Knowing this to be an orthogonal polynomial ensemble, there is still a question of what are exactly the orthogonal polynomials with respect to the weight function defined. Consider $\{P_m(z) = \sum_{j=0}^m a_j z^j\}_m^{(w)}$ a family of orthogonal polynomials with respect to w where P_m has degree m .

Lemma 3.2. *The set $\{z^j\}_j^{(w)}$ is the proper orthogonal family of polynomials in relation to the weight function $w(z) = e^{-nQ(z)}$ defined as in as in ??.*

Proof. We know by the assumption of orthogonality that

$$\int P_k(x)P_m(x)w(x)dx = \langle P_k, P_m \rangle_w = \begin{cases} 0 & \text{if } m \neq k, \\ 1 & \text{if } m = k. \end{cases}$$

Let us take some arbitrary base of the polynomial space $\{z^j\}_j^{(w)}$. We apply the Gram Schmidt scheme to get the proper orthogonality with respect to the weight w . If $P_m = x^m$, then the orthogonal element of the basis $\tilde{P}_m$ should be

$$\tilde{P}_m = P_m - \sum_{j=0}^{m-1} \frac{\langle P_m, P_j \rangle_w}{\langle P_j, P_j \rangle_w} P_j,$$

but notice that

$$\begin{aligned}
 \langle P_m, P_j \rangle_w &= \int_{\mathbb{C}} z^m \bar{z}^j e^{-nQ(z)} d^2z, \\
 &= \int_0^\infty \int_0^{2\pi} r^{m-j} e^{i\theta(m-j)} e^{-nq(r)} r dr d\theta, \\
 &= \int_0^{2\pi} e^{i\theta(m-j)} d\theta \int_0^\infty r^{m-j+1} e^{-nq(r)} dr, \\
 &= 0, \quad \text{since } m \neq j.
 \end{aligned}$$

Therefore, we have that $\{z^j\}_j^{(w)}$ is the proper orthogonal family. $\square$

We also want to determine the support of the equilibrium measure for the eigenvalues of the planar normal ensembles. For that, we make an ansatz and use ?? to confirm it.

Theorem 3.5. *Let w be as in ?. If r_0 is the largest solution of $rq'(r) = 0$ and r_1 is the smallest solution of $rq'(r) = 2$ then the support of μ_Q is given by the ring $\mathcal{S} = \{z | r_0 \leq |z| \leq r_1\}$ and $d\mu_Q$ is given by*

$$d\mu_Q(z) = \frac{1}{4\pi} (rq'(r))' \mathbb{1}_{\mathcal{S}} dr d\phi, \quad z = r e^{i\phi}.$$

The set $\mathcal{S}$ can be either an annulus (if $r_0 > 0$) or a disk (if $r_0 = 0$).

Proof. We will start on the *ansatz* that this support has two profiles; it can either be a disk centered at the origin with radius r_1 denoted by $D(r_1)$ or it can be an annulus centered at the origin with internal radius r_0 and external radius r_1 denoted by $A(r_0, r_1)$. Let $\mu \in \mathcal{M}^1(\mathbb{C})$ be a measure such that $d\mu = \frac{1}{4\pi} (rq'(r))' \mathbb{1}_{\mathcal{S}} d\phi dr$. We want that

$$\int_{\mathbb{C}} d\mu = 1,$$

that is,

$$r_1 q'(r_1) - r_0 q'(r_0) = 2.$$

Naturally, with r_0, r_1 as in the theorem, we have that this relation is satisfied. Now, by ??, if μ satisfies that

$$\begin{aligned}
 2U^\mu(z) &\geq l - Q(z), \quad z \in \mathbb{C} \\
 2U^\mu(z) &= l - Q(z), \quad z \in \{z \mid r_0 \leq |z| \leq r_1\}.
 \end{aligned}$$

with $I_Q(\mu) < \infty$, then $\mu_Q = \mu$. We first prove the formula

$$\int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi = \begin{cases} 2\pi \ln \frac{1}{r}, & |z| \leq r, \\ 2\pi \ln \frac{1}{|z|}, & |z| > r. \end{cases}$$

We start by considering the case $|z| > r$. We write

$$\ln \frac{1}{|z - r e^{i\theta}|} = \ln \frac{1}{r} + \ln \frac{1}{\left| \frac{z}{r} - e^{i\theta} \right|} = \ln \frac{1}{r} + \Re \ln \frac{1}{\omega - s}$$

where we set $s = e^{i\theta}$ and $\omega = z/r$. Then, $d\theta = ds/(is)$ and by Cauchy's theorem

$$\int_{\pi}^{-\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi = \ln \frac{1}{r} + \Re \int_{\pi}^{-\pi} \ln \frac{1}{\omega - s} \frac{1}{is} ds = -2\pi \ln(|z|).$$

In a very similar fashion, for $|z| \leq r$,

$$\ln \frac{1}{|z - r e^{i\theta}|} = \ln \frac{1}{|z|} + \ln \frac{1}{\left| e^{-i\theta} - \frac{r}{z} \right|} = \ln \frac{1}{|z|} + \Re \ln \frac{1}{s - \omega}$$

where we set $s = e^{-i\theta}$ and $\omega = r/z$. Then, $d\theta = -ds/(is)$ and by Cauchy's theorem again

$$\int_{\pi}^{-\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi = \ln \frac{1}{|z|} - \int_{\pi}^{-\pi} \ln \frac{1}{s - \omega} \frac{1}{is} ds = -2\pi \ln(r).$$

Now, with this, we calculate that

$$\begin{aligned} U^{\mu}(z) &= \frac{1}{4\pi} \int_{\mathbb{R}} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\phi}|} (rq'(r))' \mathbb{1}_S ds dr d\phi, \\ &= \frac{1}{4\pi} \int_{r_0}^{r_1} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\phi}|} (rq'(r))' dr d\phi, \\ &= \frac{1}{4\pi} \int_{r_0}^{r_1} (rq'(r))' \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi dr. \end{aligned}$$

And, then, for $|z| < r_0$

$$\begin{aligned} U^{\mu}(z) &= \frac{1}{2} \int_{r_0}^{r_1} (rq'(r))' \ln \left(\frac{1}{r} \right) dr = -\frac{1}{2} (\ln(r_1) r_1 q'(r_1) - \ln(r_0) r_0 q'(r_0) - q(r_1) + q(r_0)), \\ &= -\frac{1}{2} (2 \ln(r_1) - q(r_1)) - \frac{1}{2} q(r_0), \end{aligned}$$

for $|z| > r_1$

$$U^{\mu}(z) = \frac{1}{2} \int_{r_0}^{r_1} (rq'(r))' \ln \left(\frac{1}{z} \right) dr = \ln \frac{1}{|z|},$$

and, finally, for $r_0 \leq |z| \leq r_1$

$$\begin{aligned} U^{\mu}(z) &= \frac{1}{2} \int_{r_0}^{|z|} (rq'(r))' \ln \left(\frac{1}{r} \right) dr + \frac{1}{2} \int_{|z|}^{r_1} (rq'(r))' \ln \left(\frac{1}{z} \right) dr, \\ &= -\frac{1}{2} (2 \ln(r_1) - q(r_1)) - \frac{1}{2} q(|z|). \end{aligned}$$

Define $l = -(2 \ln(r_1) - q(r_1))$, then

$$2U^{\mu}(z) = \begin{cases} l - q(r_0), & |z| < r_0, \\ 2 \ln \frac{1}{|z|}, & |z| > r_1, \\ l - q(|z|), & r_0 \leq |z| \leq r_1. \end{cases}$$

We, as in ??, need to prove that for $z \in \mathbb{C}$, we have $2U^\mu(z) \geq l - q(|z|)$ while on $r_0 \leq |z| \leq r_1$ we have $2U^\mu(z) = l - q(|z|)$.

Firstly, for $0 < |z| < r_0$ we need that $q(r_0) - q(|z|) \geq 0$ such that $2U^\mu(z) = l - q(r_0)$ is less than or equal to $l - q(|z|)$. This is true because of the growing condition on q . Moreover, we need for $|z| > r_1$, that $-2 \ln |z| + q(|z|) \geq -2 \ln(r_1) + q(r_1)$. The same argument on the growing of q and $\ln$ holds to state that this is also true. Therefore, $\mu_Q = \mu$ and we have found our equilibrium measure. $\square$

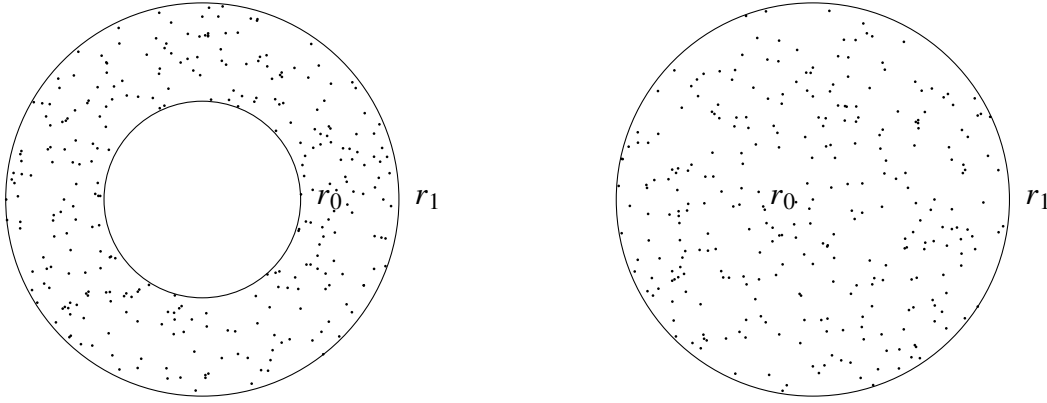


Figure 5 – The figure depicts the two topological cases for the support of the equilibrium measure - the droplet. On the left $r_0 > 0$ defining the annulus and on the right $r_0 = 0$ defining the disk.

Definition 3.6. We call **Droplet** the support of μ_Q , which in our case is the set of all complex values z such that $r_0 \leq |z| \leq r_1$.

3.3 Solving for the Free Energy

Remember the expression for the Gibbs-Boltzmann measure of Coulomb gases

$$dp_n(\chi_n) = \frac{1}{\mathcal{Z}_n^Q} e^{-\beta n^2 \mathcal{H}(\chi_n)} \prod_{i=1}^n d^d x_i.$$

As discussed, this expression refers to a thermodynamic fluid of particles with positions $\chi_n = x_1, x_2, \dots, x_n$ on a conductor Σ , in equilibrium at some inverse temperature β and submitted to a confining potential Q and a repulsive logarithmic term of interaction. Because of the term βn^2 , taking the asymptotic $n \rightarrow \infty$ means at the same time zero temperature and the thermodynamic limit. Physically, understanding the equilibrium positions would be equivalent to minimizing the Free Energy $F = -(1/\beta) \ln \mathcal{Z}_n^Q$. Let's describe the idea behind such minimization. This section is *not rigorous*, its only purpose is to connect the idea of equilibrium problems with the asymptotics we will perform in Chapter 5 and elucidate where the expression for the free energy comes from. This argument, in more detail, can be found in Livan, Novaes and Vivo (2018).

Lets first take the normalized counting measure, already introduced in ??,

$$\mathbf{n}(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - x_i).$$

Note that using this function we write the unity as

$$1 = \int d[\mathbf{n}(x)] \delta \left[\mathbf{n}(x) - \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) \right],$$

we rewrite our partition function from (??) as

$$\mathcal{Z}_n^Q = C_{n,\beta} \int d[\mathbf{n}(x)] \int_{\mathbb{R}^{dn}} \prod_{j=1}^n dx_j e^{-\beta n^2 \mathcal{H}(\chi_n)} \delta \left[\mathbf{n}(x) - \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) \right].$$

We do not want to deal with summations, in this context, integrals can be much easier to deal with. Then, we also want to convert the expression for the Hamiltonian. The confining term is easier, we write

$$\frac{1}{n} \sum_{i=1}^n Q(x) = \frac{1}{n} \int_{\mathbb{R}^d} n(x) Q(x) dx.$$

For the interaction term, one must first note that

$$\frac{1}{2n^2} \sum_{i \neq j} \ln |x_i - x_j| = \frac{1}{2n^2} \left[\sum_{i,j} \ln |x_i - x_j| - \sum_i \Delta x_i \right]$$

where we call Δx_i is the cutoff distance and refers to the term $i = j$ in the discrete case and two neighboring charges approximating in the continuous case. This should ensure that the functional will still produce a finite result. Now we write

$$\frac{1}{2n^2} \sum_{i \neq j} \ln |x_i - x_j| = \frac{1}{2n^2} n^2 \int \int_{\mathbb{R}^{2d}} dx dy n(x) n(y) \ln |x - y| - \frac{1}{2n^2} n \int_{\mathbb{R}^d} dx n(x) \ln \Delta(x).$$

and then

$$\begin{aligned} \mathcal{Z}_n^Q &= C_{n,\beta} \int d[\mathbf{n}(x)] e^{-\beta n^2 \mathcal{H}(\mathbf{n}(x))} \int_{\mathbb{R}^{dn}} \prod_{j=1}^n dx_j \delta \left[\mathbf{n}(x) - \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) \right] \\ &=: C_{n,\beta} \int d[\mathbf{n}(x)] e^{-\beta n^2 \mathcal{H}(\mathbf{n}(x))} I_n[\mathbf{n}(x)], \end{aligned}$$

where

$$\mathcal{H}(\mathbf{n}(x)) = \int_{\mathbb{R}^d} n(x) Q(x) dx + \frac{1}{2} \int \int_{\mathbb{R}^{2d}} dx dy n(x) n(y) \ln |x - y| - \frac{1}{2n} \int_{\mathbb{R}^d} dx n(x) \ln \Delta(x).$$

Physically it is possible to think of $I_n[\mathbf{n}(x)]$ as a count of how many microstates χ_n are compatible with the macrostate $\mathbf{n}(x)$. The logarithm of this quantity is proportional to the

entropy of such system. Rewrite this function as

$$\begin{aligned} I_n[\mathbf{n}(x)] &= \int d[\mathbf{n}(x)] \int_{\mathbb{R}^{dn}} \prod_{j=1}^n dx_j \exp \left[in \int dx \mathbf{n}(x) \hat{\mathbf{n}}(x) - i \int dx \hat{\mathbf{n}}(x) \sum_{i=1}^n \delta(x - x_i) \right] \\ &= \int d[\mathbf{n}(x)] \exp \left[in \int_{\mathbb{R}^{dn}} dx \mathbf{n}(x) \hat{\mathbf{n}}(x) \right] \left[\int_{\mathbb{R}^d} dy e^{i \int_{\mathbb{R}^d} dx \hat{\mathbf{n}}(x) \delta(x - x_i)} \right]^n \\ &= \int d[\hat{\mathbf{n}}(x)] e^{nS[\hat{\mathbf{n}}(x)|\mathbf{n}(x)]}, \end{aligned}$$

where

$$S[\hat{\mathbf{n}}(x)|\mathbf{n}(x)] = i \int_{\mathbb{R}^{dn}} dx \mathbf{n}(x) \hat{\mathbf{n}}(x) + \ln \int_{\mathbb{R}^d} dy e^{-i\hat{\mathbf{n}}(y)}.$$

Notice here that (??) is of a similar form from the one required by the Laplace asymptotic technique discussed in [Chapter 4](#). We estimate it, if S has a maxima with strictly negative second derivative, by the evaluation at its maximum point. We find

$$\begin{aligned} 0 &= \frac{\partial S}{\partial \hat{\mathbf{n}}(x)} = in(x) - i \frac{e^{-i\hat{\mathbf{n}}(x)}}{\int_{\mathbb{R}^d} dy e^{-i\hat{\mathbf{n}}(y)}} \\ \implies -i\hat{\mathbf{n}}(x) &= -\ln(\mathbf{n}(x)) - \ln \int_{\mathbb{R}^d} dy e^{-i\hat{\mathbf{n}}(y)}, \end{aligned}$$

for the critical counting function. Then, substituting,

$$I_n[\mathbf{n}(x)] \approx \exp \left[-n \int_{\mathbb{R}^{nd}} dx \mathbf{n}(x) \ln(\mathbf{n}(x)) \right],$$

which has the form of the Shannon entropy of the density $\mathbf{n}(x)$. There is a classical argument that states that $\Delta(x) \approx c/(n\mathbf{n}(x))$, i.e., the higher the density of particles around a point x , the smaller the average distance between them. Thus,

$$\begin{aligned} \int_{\mathbb{R}^d} dx \mathbf{n}(x) \ln \Delta(x) &\approx \int_{\mathbb{R}^d} dx \mathbf{n}(x) \ln \left(\frac{c}{n\mathbf{n}(x)} \right) \\ &= \int_{\mathbb{R}^d} dx \mathbf{n}(x) (\ln c - \ln n + \ln \mathbf{n}(x)) \\ &= \int_{\mathbb{R}^d} dx \mathbf{n}(x) \ln c - \int_{\mathbb{R}^d} dx \mathbf{n}(x) \ln n + \int_{\mathbb{R}^d} dx \mathbf{n}(x) \ln \mathbf{n}(x). \end{aligned}$$

Return now to the partition function and use the relations found to write

$$\begin{aligned} \mathcal{Z}_n^Q &\approx C_{n,\beta} \int d[\mathbf{n}(x)] \exp \left[-\beta n^2 \mathcal{H}(\mathbf{n}(x)) \right] \exp \left[-n \int_{\mathbb{R}^{nd}} dx \mathbf{n}(x) \ln(\mathbf{n}(x)) \right], \\ &\approx C_{n,\beta} \int d[\mathbf{n}(x)] \exp \left[-\beta n^2 \mathfrak{F}_0[\mathbf{n}(x)] + \frac{\beta n}{2} \mathfrak{F}_1[\mathbf{n}(x)] + \frac{\beta}{2} n \ln(n) - \frac{\beta}{2} n \ln(c) \right] \exp \left[-n \mathfrak{F}_1[\mathbf{n}(x)] \right], \\ &\approx C_{n,\beta} \int d[\mathbf{n}(x)] \exp \left[-\beta n^2 \mathfrak{F}_0[\mathbf{n}(x)] + \frac{\beta}{2} n \ln(n) + n \left(\frac{\beta}{2} - 1 \right) \mathfrak{F}_1[\mathbf{n}(x)] - \frac{\beta}{2} n \ln(c) \right], \end{aligned}$$

where

$$\begin{aligned} \mathfrak{F}_0[\mathbf{n}(x)] &:= \int_{\mathbb{R}^d} \mathbf{n}(x) Q(x) dx + \frac{1}{2} \int \int_{\mathbb{R}^{2d}} dx dy \mathbf{n}(x) \mathbf{n}(y) \ln |x - y|, \\ \mathfrak{F}_1[\mathbf{n}(x)] &:= \int_{\mathbb{R}^d} dx \mathbf{n}(x) \ln \mathbf{n}(x). \end{aligned}$$

Note that, now, one again perform a saddle point estimation for the asymptotic in a critical point $\mathbf{n}^*(x)$ to get

$$\mathcal{Z}_n^Q \approx \exp \left[-\beta n^2 \mathfrak{F}_0[\mathbf{n}^*(x)] + \frac{\beta}{2} n \ln(n) + n \left(\frac{\beta}{2} - 1 \right) \mathfrak{F}_1[\mathbf{n}^*(x)] - \frac{\beta}{2} n \ln(c) \right]$$

and finally for the Free Energy

$$\ln(\mathcal{Z}_n^Q) = -\beta n^2 \mathfrak{F}_0[\mathbf{n}^*(x)] + \frac{\beta}{2} n \ln(n) + n \left(\frac{\beta}{2} - 1 \right) \mathfrak{F}_1[\mathbf{n}^*(x)] - \frac{\beta}{2} n \ln(c) + \mathcal{O}(n).$$

Obtaining a expression for the Free Energy when $\beta = 2$, rigorously, is the main goal for [Chapter 5](#).

ASYMPTOTICS

FREE ENERGY

We are going to discuss in this chapter the ideas developed in the paper [Byun, Kang and Seo \(2023\)](#), where asymptotics of partition functions were developed for a large class of random matrix ensembles taking advantage of the orthogonal polynomials present in these models. By the end, we hope to clarify with mathematical rigor the origin for the limit-behavior expression for the *Partition Function* $\mathcal{Z}_n^Q$ given by

$$\ln(\mathcal{Z}_n^Q) = -I_Q n^2 + \frac{1}{2} n \ln n + E_Q n + G \ln n + F^Q + o(1). \quad (5.1)$$

Starting from the p.d.f. of the ensemble, one can often express the j.p.d.f. of the set of eigenvalues that arise from realizations of random matrices. We have seen in ?? that these j.p.d.f. define point process - a measure on the space of configuration of eigenvalues. Partition functions perform the role of the normalization for the probability measure on j.p.d.f. of eigenvalues. In this instance, we deal with particles in the complex plane that come from *random normal matrix ensembles* as defined in ?. We consider here confining potentials Q that are radially symmetric as specified in ?. For a large class of such Q , the first term I_Q in (5.1) was known for a long time, given by the system's energy when distributed according to the equilibrium measure - a distribution that minimizes the system's energy. Note that this quantity was discussed in ?? when discussing the determination of the equilibrium measure. The coefficient $1/2$ of the order $n \ln n$, as well as the entropy term E_Q , were described in the work of [Leblè and Serfaty \(2017\)](#), also for rather large classes of Q . The remaining terms in this expansion appeared firstly at [Byun, Kang and Seo \(2023\)](#). The term G appears to depend only on the genus of the limiting spectrum - whether it is an annulus or a disk - which, surprisingly, seems to not have been observed in the previous literature. Although some recent work, such as in [Hedenmalm and Wennman \(2020\)](#), may indicate a direction to expand some of these results to larger classes of potentials, the symmetry of the external potential is, so far, still a necessary condition for the techniques presented henceforth.

5.1 Outline of the Technique

Given some measure $p_n(z_1, \dots, z_n)$, one can think of its norming function. This is what is called here the partition function $\mathcal{Z}_n^Q$. Its logarithm represents the free energy of the system and is a quantity of interest in physics and mathematics. To get its asymptotics we will perform the following steps.

- *Partition functions can be expressed in terms of the norming constants:* Remind ?? and (??) that stated that normal ensembles' eigenvalues univariate density can be expressed as a Kernel; a function of the orthogonal polynomial with respect to the weight function w . An import result for normal ensembles gives the partition function as a function of the product of norming constants $h_j := h_{j,n}$. Then, as in ??, we can write the free energy as

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

- *Asymptotics of norming constant using the asymptotic Laplace technique:* If we set the weight function as $w(z) = e^{-nQ(z)}$ for Q a radially symmetric potential, i.e. $Q(z) = q(|z|)$, the integral expression for the norming constants can be rewritten as a real line integral and approximated by the Laplace technique.

$$\begin{aligned} h_j &= \int_{\mathbb{C}} |z|^{2j} e^{-nQ(z)} d^2 z, \\ &= \int_{\mathbb{R}_+} 2r e^{-n\left(q(r) - \frac{2j}{n} \ln(r)\right)} dr, \\ &= \int_{\mathbb{R}_+} 2r e^{-nV_{\tau(j)}(r)} dr \approx e^{-nV_{\tau(j)}(r_c(\tau(j)))}. \end{aligned}$$

- *Asymptotic of the summation on $\ln h_j$ using quadrature:* Finally, to approximate the summation on $\ln(h_j)$ we use the previously discussed idea of quadrature in ?? relating the limit summation with the corresponding integral

$$-\sum_{j=0}^{n-1} \ln h_j \approx -n^2 \int_{r_0}^{r_1} V(r) \frac{(rq'(r))'}{2} dr = -n^2 I_Q[\mu_Q].$$

This technique can take us even further in order for the asymptotic expression. Above, we only used first order approximations, however in the following sections we show that doing the detailed computations, we can get all the terms down to the constant order for the asymptotic expansion. For the next section however we expand on this simplified version to give an outline for the more complex calculations. Modulo minor adaptations, the results we state for the random planar normal matrix ensemble are also valid in the context of the *planar symplectic ensemble* - defined with an additional complex conjugation symmetry. We however omit this case to focus on clarity for the technique.

5.1.1 Norming Constants

We must study the asymptotic behavior of the norming constant h_j described in ???. We expand on the idea in Section 5.1 where we stated that the integral could be approximated by its value in a critical point. This is, of course, based on the Laplace technique described in ??. Defining $z = r e^{i\theta}$ ($|z| = r$), because of the symmetry on Q we write

$$\begin{aligned} h_j &= \int_{\mathbb{C}} |z|^{2j} e^{-nQ(z)} d^2z \\ &= \int_{\Theta} \int_{\mathbb{R}_+} r^{2j} e^{-nq(r)} \frac{r}{\pi} dr d\theta, \\ &= \frac{2\pi}{\pi} \int_{\mathbb{R}_+} r^{2j+1} e^{-nq(r)} dr, \\ &= \int_{\mathbb{R}_+} 2r e^{-n\left(q(r) - \frac{2j}{n} \ln(r)\right)} dr. \end{aligned}$$

We redefine our potential using

$$V_{\tau(j)} = q(r) - 2\tau(j) \ln(r),$$

where $\tau(j) := j/n \in [0, 1)$. Then, rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r e^{-nV_{\tau(j)}(r)} dr.$$

Note that now τ will replace the role of the index j marking the “degree” of the norming constant. Taking a look at the first derivative of the newly defined potential we find that the critical point r_c must satisfy

$$V'_{\tau}(r) = q'(r) - \frac{2\tau}{r} \implies r_c q'(r_c) = 2\tau.$$

We found a critical point r_c of the potential, parameterized by the degree of the polynomial we are calculating (given by τ). We will denote $r_{\tau(j)}$ the r such that $r q'(r) = 2\tau(j)$, that is, for each τ , we denoted $r_{\tau(j)}$ its critical point. Of course, because we know the range of τ , it is worth defining the limit points when $\tau = 0$ and $\tau = 1$ as

$$\begin{cases} r_0, & \text{the largest solution of } r q'(r) = 0, \\ r_1, & \text{the smallest solution of } r q'(r) = 2. \end{cases}$$

Note that these two radius, r_0 and r_1 define what we have called the Droplet in ??. They define the internal and external radius of the border for the equilibrium measure of the planar normal ensembles. Our main goal here is to evaluate the integral using the surroundings of the critical point. If this point is unique, then we only need to calculate the integral in a single small interval around $r_c(\tau(j)) := r_{\tau(j)}$. This is, of course, true: We mentioned that the properties on the potential directly implies that $r q'(r)$ is a strictly increasing function of $[r_0, r_1]$, then the solution must be unique.

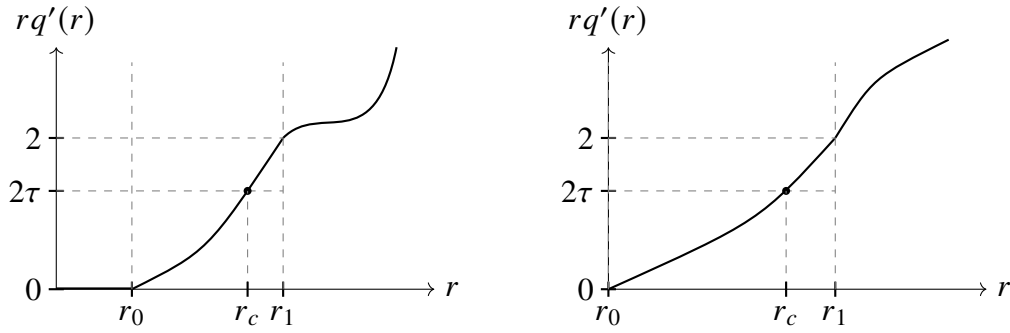


Figure 6 – Graphs for the two cases of the function $rq'(r)$. On the left, we show the case for $r_0 > 0$, corresponding to the annulus droplet. On the right, we show the case for $r_0 = 0$, corresponding to the disk droplet.

Now, as outside the droplet the growth is not strict, the definitions of r_0 and r_1 had to be as mentioned before: Respectively the largest and smallest solution to the proper instance of $rq'(r) = 2\tau$. An important distinction must be made between the cases when $r_0 = 0$ and $r_0 > 0$, corresponding to [Section 5.3](#) and [Section 5.2](#), respectively. We note that $r_\tau = r(\tau)$ defined such that $r_\tau q'(r_\tau) = 2\tau$ is an increasing function of τ and that $r_0 \leq r_\tau \leq r_1 \forall \tau \in [0, 1)$. Whichever the case, we show that the integral is approximated by its value on the critical point as

$$\begin{aligned} h_j &= \int_{\mathbb{R}_+} 2r e^{-nV_{\tau(j)}(r)} dr, \\ &\approx e^{-nV_{\tau(j)}(r_c(\tau(j)))}. \end{aligned} \quad (5.2)$$

5.1.2 Approximating the Summation

Now, we remember that the expression for the free energy, accordingly to ?? is given in terms of the norming constants by

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

Having the asymptotic expression (5.2) for the norming constant, we now want to estimate the summation on the logarithm of such expression. We write

$$\ln \mathcal{Z}_n^Q = \ln n! - n \sum_{j=0}^{n-1} V_{\tau(j)}(r_c(\tau(j))).$$

Performing this sum is not a clear procedure at first. One way to do it is to reconsider the quadrature idea described in ??. With this, we can rewrite our summation as an integral, which may be easier to solve. In our case, the summation can be approximated as

$$n \sum_{j=0}^{n-1} V_{\tau(j)}(r_c(\tau(j))) \approx n^2 \int_0^1 V_\tau(r_c(\tau)) d\tau$$

We remember the expression for the derivative of the potential $V'_\tau(r) = q'(r) - 2\tau/r$. Note that the critical point $r_{\tau(j)}$ satisfies $r_{\tau(j)}q'(r_{\tau(j)}) = 2\tau(j)$.

Lemma 5.1. *The differential of r as a function of τ at $r(\tau)$ is given by*

$$\frac{dr(\tau)}{d\tau} = \frac{2}{r(\tau)V_\tau^{(2)}(r(\tau))} = \frac{2}{(rq'(r))'} \Big|_{r=r(\tau)}.$$

where $V^{(2)}$ is the second derivative of V .

Proof. Note the differential

$$\frac{dr(\tau)}{d\tau}q'(r) + \frac{dr(\tau)}{d\tau}rq''(r) = 2$$

which leads to

$$\begin{aligned} \frac{dr(\tau)}{d\tau} &= \frac{2}{q'(r) + rq''(r)}, \\ &= \frac{2}{(rq'(r))'}. \end{aligned}$$

Evaluating at $r(\tau)$ we have

$$\frac{2}{(rq'(r))'} \Big|_{r=r(\tau)} = \frac{2}{q'(r(\tau)) + r(\tau)q''(r(\tau))} = \frac{2}{\frac{2\tau}{r(\tau)} + r(\tau)q''(r(\tau))} = \frac{2}{r(\tau)V_\tau^{(2)}(r(\tau))}.$$

□

With [Lemma 5.1](#), we write the integral in terms of r as

$$n \sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) \approx n^2 \int_{r_0}^{r_1} V(r) \frac{(rq'(r))'}{2} dr.$$

This lead us to the main term in the expression (5.1), with order n^2 . Note that this does not depend on the cases for r_0 bigger or equal to 0. This difference only appears in lower order approximations. This can be seen if one considers a more complete approximation and write

$$\begin{aligned} n \sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) &= n^2 \int_{r_0}^{r_1} V(r) \frac{(rq'(r))'}{2} dr \\ &\quad - \frac{n}{2} (V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(0)}(r_{\tau(0)})) \\ &\quad + \frac{n}{12} [\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0}] \\ &\quad + O(n^{-3}). \end{aligned}$$

Namely, the dependency on the genus of the droplet is expressed on the coefficient of $\ln n$ order and changes from the disk to the annulus case. This general first-term approach does not have such detail. We will, in the following section, derive the same asymptotic using very similar techniques. However, we will show that this approach is capable of capturing even constant terms if done more carefully.

5.2 Annulus

We start by considering the annulus regime, when the droplet has $r_0 > 0$. When we say that $r_0 > 0$, we are saying that the equilibrium measure for model introduced in ?? is non-zero in the region of space equivalent to the set of points z such that $|z| \in [r_0, r_1]$. Our objective with this section is to deduce the exact expression, analogous to (5.1), of the asymptotic of the free energy $\ln \mathcal{Z}_n^Q$ when $r_0 > 0$. For that, we will state the result and then follow the technique outlined in Section 5.1. We now state the main theorem of the section regarding the asymptotic for the partition function.

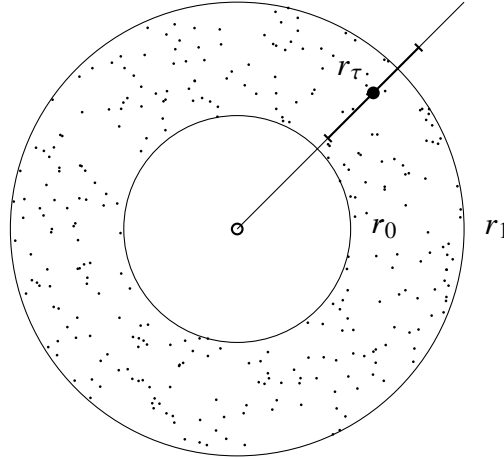


Figure 7 – The figure depicts an annulus referring to the equilibrium measure when $r_0 > 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for.

Theorem 5.1. *For the annulus case, where $r_0 > 0$, we have the following expression for the free energy asymptotically on n given by*

$$\begin{aligned} \ln \mathcal{Z}_n^Q = & -n^2 I_Q[\mu_Q] + \frac{n}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{n}{2} \ln(n) \\ & + \frac{\ln(2\pi)}{2} + \frac{1}{2} \ln(n) + F_V[A_{[r_0, r_1]}] + O(n^{-1}) \end{aligned}$$

where we set

$$\begin{aligned} F_V[A_{[r_0, r_1]}] := & -\frac{2}{3} \ln\left(\frac{r_1}{r_0}\right) + \frac{5}{32} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) \\ & - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr, \\ I_Q[\mu_Q] := & \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right], \end{aligned}$$

and

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right) (sq'(s))' ds.$$

We start by studying the asymptotic for the norming constants h_j . Then, we derive the formulas for the summation on its logarithm $\ln(h_j)$. These results should be enough to prove the previous theorem. The following subsections are referring to these two results.

5.2.1 Asymptotics of the Norming Constant

For this subsection, we will focus on getting the asymptotic for the integral

$$h_j = \int_{\mathbb{R}_+} 2r e^{-nV_{\tau(j)}(r)} dr$$

for each of the degrees j . With this, we can proceed to examine the summation over h_j needed to evaluate the free energy as it was shown in ??.

Theorem 5.2. *We have the following asymptotic of n for h_j , with $0 \leq j \leq n-1$,*

$$h_j = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + O(n^{-2}) \right], \quad (5.3)$$

where we have set

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24}.$$

The proof for such theorem passes through [Lemma 5.2](#) and [Lemma 5.3](#). We begin as usual when applying asymptotic techniques in integrals; separating a central part where the integral value is concentrated and another part where we control the order of contribution. This is possible because the integrand is proportional to some exponential, with negative exponent, with a unique critical point, around which, the value of the expression concentrates, as discussed in [Chapter 4](#). Define $\delta_n = \ln(n)/\sqrt{n}$ - this will be the region around the critical point which contribution we will consider - then, we have

$$h_j = \int_{|r-r_j| > \delta_n} 2r e^{-nV_{\tau(j)}(r)} dr + \int_{|r-r_j| < \delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \quad (5.4)$$

As mentioned, the first integral must be somehow controlled and the second estimated as the main value for the result. We begin by showing that the first term has an error contribution.

Lemma 5.2. *We have the expression for the integral with $|r - r_j| > \delta_n$ as in (5.4) and $0 \leq j \leq n-1$, asymptotically on n , given by*

$$\int_{|r-r_j| > \delta_n} 2r e^{-nV_{\tau(j)}(r)} dr = e^{-nV_{\tau(j)}(r_{\tau(j)})} \epsilon_n$$

where $\epsilon_n = O(e^{-c(\ln(n))^2})$.

Proof. Let us first show that we can control the integral, for that, rewrite

$$\begin{aligned}
& \int_{|r-r_j|>\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\
&= e^{-nV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_n} 2r e^{-n(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\
&= e^{-nV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_n} 2r e^{-n(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(M-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\
&= e^{-nV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_n} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(n-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr.
\end{aligned}$$

Now, we need some way to estimate the difference $(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)}))$. If we are able to say that this difference is at least $O(n^{-1})$, we control the growth of the integral choosing M to be large enough such that

$$\int_{|r-r_j|>\delta_n} \left(e^{-q(r)} r^{2\tau(j)} \right)^M r dr$$

converges. Using the definition of the function V and the mean value theorem, we start by evaluating the difference $\Delta(j) = V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)})$ in the following manner

$$\begin{aligned}
\Delta(j) &= q(r_{\tau(j+1)}) - q(r_{\tau(j)}) - 2(\tau(j+1) \ln(r_{\tau(j+1)}) - \tau(j) \ln(r_{\tau(j)})), \\
&= q'(c_1)(r_{\tau(j+1)} - r_{\tau(j)}) - \frac{2}{n} \left(\ln(r_{\tau(j+1)}) + j \ln\left(\frac{r_{\tau(j+1)}}{r_{\tau(j)}}\right) \right), \\
&= q'(c_1)r'(c_2)(\tau(j+1) - \tau(j)) - \frac{2}{n} \left(\ln(r_{\tau(j+1)}) + jO(n^{-1}) \right), \\
&= q'(c_1)r'(c_2)O(n^{-1}) - O(n^{-1}), \\
&= O(n^{-1}).
\end{aligned}$$

The last equality holds as both functions are continuous and c_1, c_2 are values that are limited; in a compact interval of the real line. Then, we add zero and use the estimation

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j+1)}(r) + V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = O(n^{-1})$$

to derive the result we are after. It is enough to estimate $V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)})$. For that, because $rq'(r)$ is increasing, for all $r > r_{\tau(j)} + \delta_n$ we say that

$$\begin{aligned}
V_{\tau(j+1)}(r) &\geq V_{\tau(j+1)}(r_{\tau(j)} + \delta_n), \\
&= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_n - r_{\tau(j+1)})^2 + O(\delta_n^3),
\end{aligned}$$

and therefore write

$$\begin{aligned}
V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) &= \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_n - r_{\tau(j+1)})^2 + O(\delta_n^3), \\
&= \delta_n^2 c + O(n^{-1}) + O(\delta_n^3), \\
&\geq \delta_n^2 c.
\end{aligned}$$

Finally, using the above,

$$V_{\tau(j+1)}(r) - V_{\tau(j+1)}(r_{\tau(j+1)}) \geq c\delta_n^2.$$

Getting this inequality back to the integral expression end up with

$$\begin{aligned} & \int_{|r-r_j|>\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\ &= e^{-nV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_n} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(n-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &\leq e^{-nV_{\tau(j)}(r_{\tau(j)})} e^{-c\delta_n^2(n-M)} \int_{|r-r_j|>\delta_n} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \end{aligned}$$

where one can check that

$$\begin{aligned} & \int_{|r-r_j|>\delta_n} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr \\ &= \int_{|r-r_j|>\delta_n} 2r e^{-M((q(r)-2\tau(j)\ln(r))-(q(r_{\tau(j)})-2\tau(j)\ln(r_{\tau(j)})))} dr, \\ &= 2 \left(e^{-q(r_{\tau(j)})} r_{\tau(j)}^{-2\tau(j)} \right)^M \int_{|r-r_j|>\delta_n} \left(e^{-q(r)} r^{2\tau(j)} \right)^M r dr. \end{aligned}$$

By assumption, we chose M to be a big enough number such that the integral above converges.

If that is true,

$$\int_{|r-r_j|>\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr = \int_{|r-r_j|>\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \leq e^{-nV_{\tau(j)}(r_{\tau(j)})} \epsilon_n,$$

with $\epsilon_n = O(e^{-c(\ln(n))^2})$. □

We now move on to the value-contributing part of the integral. The other order term should be absorbed in this one if [Theorem 5.2](#) is to be right.

Lemma 5.3. *For the second part of the integral we have, for the integral with $|r - r_j| > \delta_n$ as in (5.4) and $0 \leq j \leq n-1$, asymptotically on n ,*

$$\int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + O(n^{-2}) \right] \quad (5.5)$$

where $\mathfrak{B}(r_{\tau(j)})$ is as in [Theorem 5.2](#).

Proof. As we are dealing with a symmetric (in relation to the critical point) integral, we might

as well set a centralized variable $x = r - r_{\tau}$. With this, the integral becomes

$$\begin{aligned} & \int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\ &= \int_{-\delta_n}^{\delta_n} 2(x + r_{\tau(j)}) e^{-nV_{\tau(j)}(x+r_{\tau(j)})} dx, \\ &= e^{-nV_{\tau(j)}(r_{\tau(j)})} \int_{-\delta_n}^{\delta_n} 2(x + r_{\tau(j)}) e^{-n[V_{\tau(j)}(x+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})]} dx. \end{aligned}$$

Our interval grows with $\sqrt{n}/\ln(n)$. By re-scaling the reference variable using $y = \sqrt{n}x$ we get

$$\begin{aligned} & \int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\ &= \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \int_{-\ln(n)}^{\ln(n)} 2\left(\frac{y}{\sqrt{n}} + r_{\tau(j)}\right) e^{-n[V_{\tau(j)}(\frac{y}{\sqrt{n}}+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})]} dy. \end{aligned}$$

Using Taylor expansion for $V_{\tau(j)}\left(\frac{y}{\sqrt{n}} + r_{\tau(j)}\right)$ near the critical point $r_{\tau(j)}$ to rewrite

$$\begin{aligned} & \int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\ &= \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \int_{-\ln(n)}^{\ln(n)} 2\left(\frac{y}{\sqrt{n}} + r_{\tau(j)}\right) e^{-n\left(\frac{y^2}{2n}V_{\tau(j)}^{(2)}(r_{\tau(j)}) + \frac{y^3}{3!n^{\frac{3}{2}}}V_{\tau(j)}^{(3)}(r_{\tau(j)}) + \frac{y^4}{4!n^2}V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\left(\frac{y}{\sqrt{n}}\right)^5\right)\right)} dy, \\ &= \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{n}} \int_{-\ln(n)}^{\ln(n)} \left(\frac{y}{r_{\tau(j)}\sqrt{n}} + 1\right) e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)}) - \frac{y^3}{3!\sqrt{n}}V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!n}V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{n^{\frac{3}{2}}}\right)} dy, \\ &= \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{n}} \int_{-\ln(n)}^{\ln(n)} \left(\frac{y}{r_{\tau(j)}\sqrt{n}} + 1\right) e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} \left(e^{-\frac{y^3}{3!\sqrt{n}}V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!n}V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{n^{\frac{3}{2}}}\right)}\right) dy. \end{aligned}$$

Now we take the Taylor Series for

$$\begin{aligned} & e^{-\frac{y^3}{3!\sqrt{n}}V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!n}V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{n^{\frac{3}{2}}}\right)} = 1 - \left[\frac{y^3}{3!\sqrt{n}}V_{\tau(j)}^{(3)}(r_{\tau(j)}) + \frac{y^4}{4!n}V_{\tau(j)}^{(4)}(r_{\tau(j)}) + O\left(\frac{y^5}{n^{\frac{3}{2}}}\right) \right] \\ & - \frac{1}{2} \left[\frac{y^6}{n} \left(\frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6} \right)^2 + \frac{y^8}{n^2} \left(\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24} \right)^2 + \frac{y^7}{n^3} \left(\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{6} \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{24} \right) + O\left(\frac{y^8}{n^2}\right) \right]. \end{aligned}$$

To get the asymptotic expansion (since odd terms vanish) we reorganize the terms in the integral again to obtain

$$\begin{aligned} & \int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\ &= 2r_{\tau(j)} \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \int_{-\ln(n)}^{\ln(n)} e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} \left[1 - \frac{y^4}{3!r_{\tau(j)}n}V_{\tau(j)}^{(3)}(r_{\tau(j)}) \right. \\ & \quad \left. - \frac{y^4}{4!n}V_{\tau(j)}^{(4)}(r_{\tau(j)}) - \frac{y^6}{2 \cdot (3!)^2n}V_{\tau(j)}^{(3)}(r_{\tau(j)})^2 + O\left(\frac{y^6}{n^2}\right) \right] dy, \quad (5.6) \end{aligned}$$

then, distributing the integral,

$$\begin{aligned} & \int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\ &= 2r_{\tau(j)} \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24n} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}n} \right) \int_{-\ln(n)}^{\ln(n)} e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^4 dy \right. \\ & \quad \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72n} \int_{-\ln(n)}^{\ln(n)} e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^6 dy + \int_{-\ln(n)}^{\ln(n)} e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} dy + O(n^{-2}) \right]. \end{aligned}$$

At this point, the result is directly evaluated using the Gaussian integrals in [Proposition 5.1](#). So that we substitute the values to get

$$\begin{aligned} & \int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\ &= \sqrt{2\pi} r_{\tau(j)} \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24n} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}n} \right) \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}} \right. \\ & \quad \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72n} \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{7}{2}} + \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} + O(n^{-2}) \right], \end{aligned}$$

and, rearranging the terms we get our result

$$\int_{|r-r_j|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + O(n^{-2}) \right] \quad (5.7)$$

where we defined $\mathfrak{B}(r_{\tau(j)})$ as

$$\mathfrak{B}(r_{\tau(j)}) = -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24}.$$

Notice that the error term of [Lemma 5.2](#) is absorbed into the error order of (5.7). Therefore we have the asymptotic expression for h_j given by (5.3) and the proof for [Theorem 5.2](#) follows directly. $\square$

5.2.2 Asymptotics of the Summation

We return to ?? and turn to the calculation of each term of $\sum_{j=0}^{n-1} \ln h_j$ using the logarithm of the expansion calculated previously in [Theorem 5.2](#)

$$\sum_{j=0}^{n-1} \left[-nV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + O(n^{-2}) \right]. \quad (5.8)$$

That is, we need to prove the following theorem.

Theorem 5.3. Asymptotically on n ,

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln n + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) + F_V[A_{[r_0, r_1]}] + \mathcal{O}(n^{-1})$$

where $F_V[A_{[r_0, r_1]}]$, $I_Q[\mu_Q]$, $E_Q[\mu_Q]$ are as in [Theorem 5.1](#).

The proof comes from gathering the terms in (5.8) with its respective expressions derived in [Lemma 5.4](#) and [Lemma 5.5](#). We go term by term. [Lemma 5.4](#) refers to the asymptotic of

$$\sum_{j=0}^{n-1} (-n V_{\tau(j)}(r_{\tau(j)})),$$

and [Lemma 5.5](#) refers to the asymptotics of both terms

$$\sum_{j=0}^{n-1} \left(\ln(2\pi r_{\tau(j)}^2) - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right).$$

Then, the final expression appears by balancing the results.

Lemma 5.4. Asymptotically on n ,

$$-n \sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = -n^2 I_Q[\mu_Q] + n U_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) + \mathcal{O}(n^{-2})$$

where we have set $I_Q[\mu_Q]$ as in [Theorem 5.1](#) and

$$-2U_{\mu_Q}(0) := \int_{r_0}^{r_1} \ln(s) (s q'(s))' ds = q(r_1) - q(r_0) + 2 \ln(r_1).$$

Proof. Using the Euler-Maclaurin formula from ??,

$$\begin{aligned} \sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) &= \int_0^n V_{\tau(t)}(r_{\tau(t)}) dt - \frac{1}{2} (V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(0)}(r_{\tau(0)})) + \\ &\quad \frac{1}{12} [\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0}] + \mathcal{O}(n^{-3}). \end{aligned} \quad (5.9)$$

The first term in (5.9) is solved doing a integral by parts. Setting $s = r_{\tau(t)}$,

$$\begin{aligned} \int_0^n V_{\tau(t)}(r_{\tau(t)}) dt &= \int_0^n (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)})) dt, \\ &= \frac{1}{2} \int_{r_0}^{r_1} (q(s) - s q'(s) \ln(s)) s V^{(2)}(s) n ds, \\ &= n \left[\frac{1}{2} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds \right], \end{aligned}$$

and finally using that $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$ we apply the integration by parts on the second term to get

$$\begin{aligned}
 \frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds &= \frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \ln(s) \frac{1}{s} (s q'(s))' ds, \\
 &= \frac{1}{2} \int_{r_0}^{r_1} s q'(s) \ln(s) (s q'(s))' ds, \\
 &= \frac{1}{4} \int_{r_0}^{r_1} ((s q'(s))^2)' \ln(s) ds, \\
 &= \frac{1}{4} (s q'(s))^2 \ln(s) \Big|_{r_0}^{r_1} - \frac{1}{4} \int_{r_0}^{r_1} (s q'(s))^2 \frac{1}{s} ds \\
 &= \ln(r_1) - \frac{1}{4} \int_{r_0}^{r_1} s q'(s)^2 ds, \\
 &= \ln(r_1) - \frac{1}{2} q(r_1) + \frac{1}{4} \int_{r_0}^{r_1} q(s) (s q'(s))' ds.
 \end{aligned}$$

Leaving us with, by definition of I_Q

$$\begin{aligned}
 \int_0^n V_{\tau(t)}(r_{\tau(t)}) dt &= n \left[\frac{1}{2} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) - \frac{1}{4} \int_{r_0}^{r_1} q(s) (s q'(s))' ds \right], \\
 &= n \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right], \\
 &=: n I_Q[\mu_Q].
 \end{aligned} \tag{5.10}$$

We also need an expression for $V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})$. This is direct noting that

$$V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)}) = V_1(r_1) - V_0(r_0) = q(r_1) - q(r_0) + 2 \ln(r_1) =: 2U_{\mu_Q}(0). \tag{5.11}$$

And finally for the remaining term we use the Leibniz rule to obtain

$$\begin{aligned}
 \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} &= \partial_t (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)}))|_{t=n}, \\
 &= \partial_\tau (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)}))|_{\tau=1} \partial_t (\tau(t))|_{t=n}, \\
 &= \frac{1}{n} (\partial_\tau (q(r_\tau)) - 2(\ln(r_\tau) + \tau \partial_\tau (\ln(r_\tau))_{\tau=1})), \\
 &= \frac{1}{n} [\partial_t (q(r_{\tau(t)}) - 2 \ln(r_{\tau(t)}))_{\tau=1} - 2 \ln(r_1)]
 \end{aligned}$$

and

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{n} [\partial_t (q(r_{\tau(t)}) - 2 \ln(r_{\tau(t)}))_{\tau=0} - 2 \ln(r_0)].$$

Since we can calculate, from $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$

$$\begin{aligned}
 \partial_\tau q(r_\tau)|_{\tau=0} &= \left(\frac{2q'(r)}{(r q'(r))'} \right)_{r_0} = 0, \\
 \partial_\tau q(r_\tau)|_{\tau=1} &= \left(\frac{2q'(r)}{(r q'(r))'} \right)_{r_1} = \frac{4}{r_1 (q'(r_1) + r_1 q''(r_1))},
 \end{aligned}$$

$$\partial_\tau(2\ln(r_\tau))|_{\tau=1} = \left(\frac{4}{r(q'(r))'} \right)_{r_1} = \frac{4}{r_1(q'(r_1) + r_1 q''(r_1))},$$

we also express the difference as

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{12n} [2(\ln(r_0) - \ln(r_1))] = \frac{1}{6n} \ln\left(\frac{r_0}{r_1}\right). \quad (5.12)$$

The proof for the summation $\sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)})$ is complete if one gathers the terms in (5.10), (5.12) and (5.11). □

Lemma 5.5. *Asymptotically on n ,*

$$\sum_{j=0}^{n-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = nE_Q[\mu_Q] - \frac{1}{2} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(n^{-1})$$

and

$$\sum_{j=0}^{n-1} \ln(2\pi r_{\tau(j)}^2) = n \ln(2\pi) + 2 \left(-nU_{\mu_Q}(0) - \frac{1}{2} \ln\left(\frac{r_1}{r_0}\right) + O(n^{-1}) \right)$$

where $U_{\mu_Q}(0)$ as in Lemma 5.4 and $E_Q[\mu_Q]$ as in Theorem 5.1.

Proof. Lets start now with the second summation, to which we apply Euler-Maclaurin again

$$\begin{aligned} \sum_{j=0}^{n-1} \ln\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right) &= \int_0^n \frac{1}{4} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)}) dt - \frac{1}{2} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + \\ &\quad \frac{1}{12} \left[\partial_t \frac{1}{4} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=n} - \partial_t \frac{1}{4} \ln V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=0} \right] + O(n^{-3}). \end{aligned}$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we have

$$\int_0^n \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) dt = \frac{n}{2} \int_{r_0}^{r_1} \ln\left(\frac{1}{4} V^{(2)}(s)\right) (sq'(s))' ds.$$

And, noticing that

$$\partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) = \partial_V \left(\ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \frac{1}{4} \partial_r \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \partial_\tau(r_{\tau(t)}) = \frac{\left(\partial_r V^{(2)} \right)(r_{\tau(t)})}{r_{\tau(t)} \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right)^2},$$

we have for the third term on the right hand side,

$$\begin{aligned} \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=n} - \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=0} &= \frac{1}{2n} \left(\frac{\left(\partial_r V^{(2)} \right)(r_1)}{r_1 \left(V_1^{(2)}(r_1) \right)^2} - \frac{\left(\partial_r V^{(2)} \right)(r_0)}{r_0 \left(V_0^{(2)}(r_0) \right)^2} \right) \\ &= O(n^{-1}). \end{aligned}$$

Therefore,

$$\begin{aligned} \sum_{j=0}^{n-1} \ln \frac{1}{4} V_{\tau(j)}^{(2)}(r_{\tau(j)}) &= \frac{n}{2} \int_{r_0}^{r_1} \ln \left(\frac{1}{4} V^{(2)}(s) \right) (sq'(s))' ds - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + O(n^{-1}), \\ &:= nE_Q[\mu_Q] - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + O(n^{-1}). \end{aligned}$$

Similarly, for the first summation, we apply Euler-Maclaurin again

$$\begin{aligned} \sum_{j=0}^{n-1} \ln(2\pi r_{\tau(j)}^2) &= \ln(2\pi) + 2 \left(\int_0^n \ln(r_{\tau(t)}) dt - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) \right. \\ &\quad \left. + \frac{1}{12} [\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0}] + O(n^{-3}) \right). \end{aligned}$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we have

$$\int_0^n \ln r_{\tau(t)} dt = \frac{n}{2} \int_{r_0}^{r_1} \ln(s) (sq'(s))' ds.$$

And, noticing that

$$\partial_t \ln r_{\tau(t)} = \partial_r (\ln(r_{\tau(t)})) \partial_\tau (r_{\tau(t)}) = \frac{2}{V_{\tau(t)}^{(2)}(r_{\tau(t)}) (r_{\tau(t)})^2},$$

we have for the third term on the right hand side,

$$\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0} = \frac{1}{2n} \left(\frac{2}{V^{(2)}(r_1) (r_1)^2} - \frac{2}{V^{(2)}(r_0) (r_0)^2} \right) = O(n^{-1}).$$

Then

$$\sum_{j=0}^{n-1} \ln(2\pi r_{\tau(j)}^2) = n \ln(2\pi) + 2 \left(\frac{n}{2} \int_{r_0}^{r_1} \ln(s) (sq'(s))' ds - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) + O(n^{-1}) \right).$$

□

For the following proof, and the analogous proof for the following section, we use the auxiliary result on Gaussian integrals.

Proposition 5.1. *The Gaussian integrals below are evaluated as follows.*

$$\int_{\mathbb{R}} e^{-2ar^2} dr = \sqrt{\frac{\pi}{2}} \frac{1}{a^{1/2}}, \quad \int_{\mathbb{R}} e^{-2ar^2} r^4 dr = \sqrt{\frac{\pi}{2}} \frac{3}{16} \frac{1}{a^{5/2}} \quad \text{and} \quad \int_{\mathbb{R}} e^{-2ar^2} r^6 dr = \sqrt{\frac{\pi}{2}} \frac{15}{64} \frac{1}{a^{7/2}}.$$

Moreover,

$$\int_{\mathbb{R}} e^{-2 \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) y^2} dy = \sqrt{\frac{\pi}{2}} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}},$$

$$\int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} y^4 = \sqrt{\frac{\pi}{2}} \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}},$$

$$\int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} y^6 = \sqrt{\frac{\pi}{2}} \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{7}{2}}.$$

We now move on to the proof of [Theorem 5.3](#).

Proof. ([Theorem 5.3](#)) We only need to notice now that by ??

$$\begin{aligned} \frac{1}{n} \sum_{j=0}^{n-1} \mathfrak{B}(r_{\tau(j)}) &= \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) (sq'(s))' ds + O(n^{-2}), \\ &= \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) s V^{(2)}(s) ds + O(n^{-2}). \end{aligned}$$

Using the definition for $\mathfrak{B}(r_{\tau(j)})$ and integrating by parts,

$$\begin{aligned} \frac{1}{n} \sum_{j=0}^{n-1} \mathfrak{B}(r_{\tau(j)}) &= \\ &= \frac{1}{4} \int_{r_0}^{r_1} \left[-\frac{V^{(4)}(r)}{V^{(2)}(r)^2} \frac{1}{8} - \frac{V^{(3)}(r)}{V^{(2)}(r)^2} \frac{1}{2r} - \frac{V^{(3)}(r)^2}{V^{(2)}(r)^3} \frac{5}{24} \right] r V^{(2)}(r) dr + O(n^{-2}), \\ &= -\frac{1}{32} \int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r dr - \frac{1}{8} \int_{r_0}^{r_1} \frac{V^{(3)}(r)}{V^{(2)}(r)} dr - \frac{5}{96} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-2}), \\ &= -\frac{1}{8} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r dr - \frac{5}{96} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-2}), \end{aligned}$$

where

$$\int_{r_0}^{r_1} \frac{V^{(4)}(r)}{V^{(2)}(r)} r dr = \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} - \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

Therefore

$$\begin{aligned} &\frac{1}{n} \sum_{j=0}^{n-1} \mathfrak{B}(r_{\tau(j)}) \\ &= -\frac{3}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-2}), \quad (5.13) \end{aligned}$$

and the theorem follows from

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j &= -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6} \ln\left(\frac{r_0}{r_1}\right) - \frac{n}{2} \ln(n) - \frac{1}{2} \left(nE_Q[\mu_Q] - \frac{1}{2} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) \right) \\ &\quad + \frac{1}{2} \left(n \ln(2\pi) + 2 \left(-nU_{\mu_Q}(0) - \frac{1}{2} \ln\left(\frac{r_1}{r_0}\right) \right) \right) - \frac{3}{32} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) \\ &\quad - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-1}) \end{aligned}$$

that reduces to

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j &= -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln(n) + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) \\ &\quad - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-1}). \end{aligned}$$

□

Proof. (Theorem 5.1) From Theorem 5.2,

$$\begin{aligned} \sum_{j=0}^{n-1} \ln(h_j) &= -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln(n) + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) \\ &\quad - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-1}). \end{aligned}$$

We get the asymptotic expansion for the Free energy using that

$$\ln(n!) = n \ln(n) - n + \frac{1}{2} \ln(n) + \frac{1}{2} \ln(2\pi) + O(n^{-1})$$

that is,

$$\begin{aligned} \ln \mathcal{Z}_n^Q &= \ln n! + \sum_{j=0}^{n-1} \ln h_j, \\ &= n \ln(n) - n + \frac{1}{2} \ln(n) + \frac{1}{2} \ln(2\pi) + \left(-n^2 I_Q[\mu_Q] - \frac{n}{2} \ln(n) + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) \right. \\ &\quad \left. - \frac{2}{3} \ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr \right) + O(n^{-1}), \\ &= -n^2 I_Q[\mu_Q] + \frac{n}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{n}{2} \ln(n) + \frac{1}{2} \ln(n) + \frac{\ln(2\pi)}{2} \\ &\quad - \frac{2}{3} \ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32} \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-1}), \\ &=: -n^2 I_Q[\mu_Q] + \frac{n}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{n}{2} \ln(n) + \frac{\ln(2\pi)}{2} + \frac{1}{2} \ln(n) + F_V[A_{[r_0, r_1]}] + O(n^{-1}), \end{aligned}$$

□

5.3 Disk

We now consider the disk regime, when the droplet has $r_0 = 0$. Again, we are saying that the equilibrium measure for model introduced in ?? is non-zero in the region of space equivalent to the set of points z such that $|z| \in [r_0, r_1]$. Our objective with this section is to deduce the exact expression, analogous to (5.1), of the asymptotic of the free energy $\ln \mathcal{Z}_n^Q$ now when $r_0 = 0$. For that, we will state the result and then follow the technique outlined in Section 5.1. Byun, Kang and Seo (2023) also shows that the expression we get is different from the expression one gets when taking the limit $r_0 \rightarrow 0$ using the expression from the case $r_0 > 0$. We now state the main theorem of the section regarding the asymptotic for the partition function.

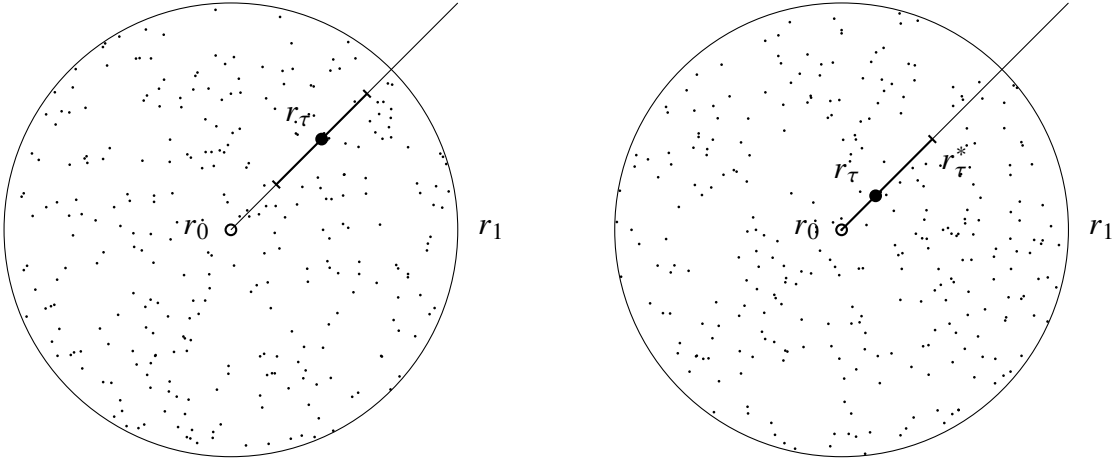


Figure 8 – The figure depicts an disk referring to the equilibrium measure when $r_0 = 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for. The left image corresponds to the case when $r_{j/n} \gg n^{-\epsilon}$ and the right image corresponds to $r_{j/n} \ll n^{-\epsilon}$. In this case r_{τ}^* will define the contributing region to the integral.

Theorem 5.4. *For the annulus case, where $r_0 = 0$, we have the following, asymptotically on n , for the free energy*

$$\begin{aligned} \ln \mathcal{Z}_n^Q = & -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) - \frac{5}{12} \ln n \frac{\ln(2\pi)}{2} \\ & + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-1/12} (\ln n)^3). \end{aligned}$$

where we have set

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (s q'(s))' ds$$

and

$$F_V[D_{r_1}] := \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - \frac{1}{32} \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12} \int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

Again, we start by studying the asymptotic for h_j and then for the summation $\sum \ln(h_j)$. This theorem then should follow from the lemmas concerning this two asymptotics.

5.3.1 Asymptotics of the Norming Constant

For this subsection, we will focus on getting the asymptotic for the integral

$$h_j = \int_{\mathbb{R}_+} 2r e^{-nV_{\tau(j)}(r)} dr$$

for each of the degrees j . With this, we can proceed to examine the summation over h_j needed to evaluate the free energy. For this case however there is one complication. One should distinguish the following two cases depending on a small constant $1/5 > \epsilon > 0$.

- Case 1: $r_{j/n} \gg n^{-\epsilon}$. For the annular droplet case where $r_0 > 0$, this case covers all $j = 0, 1, \dots, n-1$. On the other hand, for the disk droplet case where $r_0 = 0$, this case covers only $j = m_n, m_n + 1, \dots, n-1$ for $m_n = n^\epsilon$ with some $\epsilon > 0$;
- Case 2: $r_{j/n} \ll n^{-\epsilon}$. This covers the remaining disk droplet case with $j = 0, 1, \dots, m_n - 1$. Notably, the asymptotic expansion involves gamma functions in this case.

We separate this in two proofs, the first one will be regarding $j = m_n, m_n + 1, \dots, n-1$. One might assume, and would be mainly correct, that nothing is that different in the disk and annulus regimes if the points are distant from r_0 . The following results are direct adaptations of the previous arguments.

Theorem 5.5. *For $n-1 \geq j \geq m_n$, asymptotically on n ,*

$$h_j = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left(1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \epsilon_{n,1} \right)$$

where

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24},$$

and

$$\epsilon_{n,1} := O(j^{-\frac{3}{2}} (\ln n)^\alpha)$$

for some $\alpha > 0$.

To actually expand on the details, we need some useful facts. We will make a lot of use of the behavior for r_τ , expressed in [Proposition 5.2](#).

Proposition 5.2. *Due to strict sub-harmonicity of Q in a neighborhood of the droplet, r_τ , defined as before, satisfies*

$$r_\tau = \left(\frac{2\tau}{q''(0)} \right)^{\frac{1}{2}} + O(\tau) = \left(\frac{4\tau}{V_0^{(2)}(0)} \right)^{\frac{1}{2}} + O(\tau)$$

or more simply

$$r_\tau = O\left(\tau^{\frac{1}{2}}\right)$$

as $\tau \rightarrow 0$.

Proof. Begin defining a multivariate function

$$F(\tau, r) = rq'(r) - 2\tau$$

such that

$$\begin{cases} F(0, 0) = rq'(r) - 2\tau = 0, \\ \partial_r F(\tau, r)|_{(0,0)} = q'(r) + q''(r) \cdot r|_{(0,0)} = 0 \end{cases}$$

since q has critical point at $r_\tau = 0$. This means that the assumptions in the Implicit Function ?? do not hold. We must use the *blow-up* technique. Consider an auxiliary variable $v := v(r)$ such that $v(r)\tau = r^2$ and look for solutions of the redefined

$$F(\sqrt{v(r)\tau}, \tau) := F'(v, \tau) = \frac{\sqrt{v\tau} \cdot q'(\sqrt{v\tau})}{2\tau} - 1$$

around $(v_0, 0)$, where v_0 is such that $F'(v_0, 0) = 0$. Note that, expanding $q'(r)$ in Taylor around $r = 0$ give us

$$q'(\sqrt{v\tau}) = q''(0) \cdot \sqrt{v\tau} + O(v\tau),$$

which, applied in the previous function,

$$F'(v, \tau) = \frac{q''(0)}{2} v - 1 + O(v^{3/2}\tau).$$

Now, we apply the Implicit Function Theorem on F' noting that the point v_0 must satisfy

$$v_0 = \frac{2}{q''(0)}$$

and that $q''(0) > 0$ by assumption of the global minimum. Then

$$\partial_v F'(v, \tau)|_{(v_0,0)} = \frac{q''(0)}{2} > 0.$$

Then, the theorem states that there is a bijective function $v(r\tau) \in C^2$ defined in a neighborhood to $\tau = 0$, for which

$$F'(v, \tau) = 0 \quad \text{and} \quad v(0) = v_0.$$

However, $F'(v, \tau) = 0$ means precisely that

$$\sqrt{v\tau} \cdot q'(\sqrt{v\tau}) = 2\tau \quad \text{with} \quad r = r(\tau) = \sqrt{v(\tau)\tau}.$$

Leading us to believe that locally at v_0

$$r(\tau) = \left(\frac{2\tau}{q''(0)} \right)^{\frac{1}{2}} + O(\tau)$$

for some $\tau \rightarrow 0$. Next to zero, all higher powers in the expansion are just order τ . $\square$

To actually go on and prove the stated theorem we start as usual dividing the integral into two sections

$$\begin{aligned} h_j &= \int_0^\infty 2r^{2j+1} e^{-nq(r)} dr = \int_0^\infty 2r e^{-nV_{\tau(j)}(r)} dr, \\ &= \int_{|r-r_{\tau(j)}| < \delta_n} 2r e^{-nV_{\tau(j)}(r)} dr + \int_{|r-r_{\tau(j)}| > \delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \end{aligned}$$

Let's start with a result on the outer integral, which we should be able to control the growth of.

Lemma 5.6. *Consider the case $m_n \leq j \leq n-1$. If we choose M to be such that*

$$\int_{|r-r_j| > \delta_n} \left(e^{-q(r)} r^2 \right)^M r dr < \infty,$$

then, for the error-bound integral in which $|r - r_{\tau(j)}| > \delta_n$ we have the asymptotic for n given by

$$\int_{|r-r_{\tau(j)}| > \delta_n} 2r e^{-nV_{\tau(j)}(r)} dr = e^{-nV_{\tau(j)}(r_{\tau(j)})} O\left(e^{-c_2(\ln(n))^2}\right).$$

for some positive constant c_2 .

Proof. Note that for $d \geq 1$, we choose C_1 a constant uniformly for all τ and write

$$|V_\tau^{(d)}(r_\tau)| = |q^{(d)}(r_\tau) + (-1)^d 2\tau(d-1)! r_\tau^{-d}| \leq C_1(1 + \tau^{-\frac{d}{2}+1}).$$

Firstly, for $m_n \leq j < n$, we have

$$r_{\tau(j+1/2)} - r_{\tau(j)} = O((jn)^{-\frac{1}{2}}).$$

Since the potential is increasing in (r_τ, ∞) , the Taylor series expansion for V_τ is for all $r > r_{\tau(j)} + \delta_n$

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r_{\tau(j)} + \delta_n), \\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2} V_{\tau(j+1/2)}^{(2)}(r_{\tau(j+1/2)})(r_{\tau(j)} + \delta_n - r_{\tau(j+1/2)})^2 + O(\tau(j)^{-1/2} \delta_n^3). \end{aligned} \tag{5.14}$$

where $O(\tau(j)^{-1/2}\delta_n^3) = O(n^{-1}j^{-1/2}(\ln(n))^3)$. Similarly, since V_τ is decreasing in $(0, r_\tau)$, we have for all $r < r_{\tau(j)} - \delta_n$

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r_{\tau(j)} - \delta_n), \\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2}V_{\tau(j+1/2)}^{(2)}(r_{\tau(j+1/2)})(r_{\tau(j)} - \delta_n - r_{\tau(j+1/2)})^2 + O(\tau(j)^{-1/2}\delta_n^3). \end{aligned} \quad (5.15)$$

Using the Taylor series for V_τ again, we have

$$\begin{aligned} V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) &= V_{\tau(j)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) - \frac{1}{n} \ln r_{\tau(j+1/2)}, \\ &= O((jn)^{-1}) - \frac{1}{n} \ln r_{\tau(j+1/2)}. \end{aligned}$$

Noting that

$$V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) = [V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j+1/2)}(r)] + [V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)})]$$

one rewrite using (5.15) and (5.14),

$$V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}) = -\frac{1}{n} \ln r_{\tau(j+1/2)} + O\left(\frac{\log(n)^2}{n}\right).$$

Thus, it follows from the last observations that

$$\begin{aligned} e^{nV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_{\tau(j)}|>\delta_n} 2r^{2j+1} e^{-nq(r)} dr &= \int_{|r-r_{\tau(j)}|>\delta_n} 2e^{-n(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &= \int_{|r-r_{\tau(j)}|>\delta_n} 2e^{-(n-M)(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}))} e^{-M(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &\leq e^{-c_1(\ln(n))^2} e^{\frac{n-M}{n} \ln(r_{\tau(j+1/2)})} \int_{|r-r_{\tau(j)}|>\delta_n} 2e^{-M(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &= O(e^{-c_2(\ln(n))^2}). \end{aligned}$$

for some constants $c_1, c_2 > 0$ and M such that the integral above converges. The details taken here are very similar to the ones in the annulus case. $\square$

Lemma 5.7. Consider the case $m_n \leq j \leq n-1$. For the value-contributing integral we have the asymptotic of n given by

$$\int_{|r-r_{\tau(j)}|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left(1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \epsilon_{n,1} \right)$$

where $\mathfrak{B}(r_{\tau(j)})$ is as in Theorem 5.4 and

$$\epsilon_{n,1} := O(j^{-\frac{3}{2}}(\ln n)^\alpha)$$

for some $\alpha > 0$.

Proof. We now need to examine the integral near the critical point. That is,

$$\begin{aligned}
& \int_{|r-r_{\tau(j)}|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\
&= e^{-nV_{\tau(j)}(r_{\tau(j)})} \int_{-\delta_n}^{\delta_n} 2(r_{\tau(j)}+t) e^{-n\left(\frac{1}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})(t)^2+\frac{1}{3!}V_{\tau(j)}^{(3)}(r_{\tau(j)})(t)^3+\frac{1}{4!}V_{\tau(j)}^{(4)}(r_{\tau(j)})(t)^4+O(\tau(j)^{-\frac{3}{2}}t^5)\right)} dt, \\
&= e^{-nV_{\tau(j)}(r_{\tau(j)})} \int_{-\sqrt{n}\delta_n}^{\sqrt{n}\delta_n} 2e^{-\frac{1}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})t^2} \left(r_{\tau(j)}+\frac{t}{\sqrt{n}}\right) \left(1-\frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})t^3}{6\sqrt{n}}\right. \\
&\quad \left.-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})t^4}{24n}+\frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2t^6}{72n}+\epsilon_{n,1}\right) dt,
\end{aligned}$$

where $\epsilon_{n,1} = O(j^{-\frac{3}{2}}(\ln n)^\alpha)$ for some $\alpha > 0$. From here, the expression is directly identified with (5.6) and the techniques for getting the result are one-to-one with the ones already done in that case. then, distributing the integral,

$$\begin{aligned}
& \int_{|r-r_{\tau(j)}|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\
&= 2r_{\tau(j)} \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24n} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}n} \right) \int_{-\ln(n)}^{\ln(n)} e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^4 dy \right. \\
&\quad \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72n} \int_{-\ln(n)}^{\ln(n)} e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^6 dy + \int_{-\ln(n)}^{\ln(n)} e^{-\frac{y^2}{2}V_{\tau(j)}^{(2)}(r_{\tau(j)})} dy + \epsilon_{n,1} \right].
\end{aligned}$$

At this point, the result is directly evaluated using the Gaussian integrals in Proposition 5.1. So that we finally substitute the values to get

$$\begin{aligned}
& \int_{|r-r_{\tau(j)}|<\delta_n} 2r e^{-nV_{\tau(j)}(r)} dr \\
&= \sqrt{2\pi} r_{\tau(j)} \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24n} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}n} \right) \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}} \right. \\
&\quad \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72n} \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{7}{2}} + \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} + \epsilon_{n,1} \right],
\end{aligned}$$

and, rearranging the terms we should get our result. $\square$

Proof. (Theorem 5.5) This is now only a matter to notice that the expression derived in Lemma 5.6 is absorbed in to the error term of the expression in Lemma 5.7 and the result follows. $\square$

So far, nothing is considerably different from both cases as we are dealing with point distant from the origin. We must now deal with the more delicate case, for $j \leq m_n$. The behavior for h_j , in this regimen, is expressed by the following theorem.

Theorem 5.6. For $0 \leq j < m_n$, we will have the following asymptotic for the norming constant

$$h_j = e^{-nq(0)} \left[n^{-(j+1)} \Gamma(j+1) \left(\frac{1}{2} q''(0) \right)^{-(j+1)} \left(1 + O(n^{-\frac{1}{2}} (j+1/2)^{\frac{3}{2}} (\ln n)^3) \right) + \epsilon_n(j) \right]$$

where Γ is the usual Gamma function and

$$\epsilon_n(j) := O \left(e^{-c_2(j+1)(\ln n)^2} \left(\frac{j+1}{n} (\ln(n))^2 \right)^{(j+1) \frac{n-M}{n}} \right)$$

with $c_2 > 0$ a constant.

Proof. For $j < m_n$, we need to define some $r_\tau^* := r_\tau \ln(n)$ from which we separate the integral in a important part close to the origin and another part which has a lesser order. Consider then the decomposition

$$h_j = \int_0^\infty 2r^{2j+1} e^{-nq(r)} dr = \int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-nq(r)} dr + \int_{r_{\tau(j+1)}^*}^\infty 2r^{2j+1} e^{-nq(r)} dr.$$

Due to strict sub-harmonicity of Q in a neighborhood of the droplet, r_τ satisfies $r_\tau = O(\tau^{\frac{1}{2}})$ as $\tau \rightarrow 0$, this is given by [Proposition 5.2](#). Since V_τ has a global minimum at $r = r_\tau$ and increases for larger values, for all $r > r_{\tau(j+1)}^*$ we have

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1)}(r)(r_{\tau(j+1)}^*) = q(r_{\tau(j+1)}^*) - \frac{2j+1}{n} \ln(r_{\tau(j+1)}^*), \\ &= q(0) + \frac{1}{2} q''(0) (r_{\tau(j+1)}^*)^2 - \frac{2j+1}{n} \ln(r_{\tau(j+1)}^*) + O(r_{\tau(j+1)}^*)^3. \end{aligned}$$

Note that $q'(0) = 0$ since $\Delta Q(z) \in (0, \infty)$ near the origin and $\Delta Q(z) \propto q'(0)/r + q''(0)$. Take

$$\begin{aligned} e^{-nq(0)} \int_{r_{\tau(j+1)}^*}^\infty 2r^{2j+1} e^{-nq(r)+nq(0)} dr &= e^{-nq(0)} \int_{r_{\tau(j+1)}^*}^\infty 2e^{(2j+1)\ln(r)-nq(r)+nq(0)} dr, \\ &= e^{-nq(0)} \int_{r_{\tau(j+1)}^*}^\infty 2e^{-n(-\frac{2}{n}(j+\frac{1}{2})\ln(r)-q(r)+q(0))} dr, \\ &= \int_{r_{\tau(j+1)}^*}^\infty 2e^{-n(V_{\tau(j+1/2)}(r)-q(0))} dr \end{aligned}$$

and, using the inequality explicited above

$$\begin{aligned} e^{-nV_{\tau(j+1)}(r)+nq(0)} &= e^{-nV_{\tau(j+1)}(r)+nq(0)+MV_{\tau(j+1)}(r)-MV_{\tau(j+1)}(r)}, \\ &= e^{-(n-M)V_{\tau(j+1)}(r)+nq(0)-MV_{\tau(j+1)}(r)}, \\ &\leq e^{-(n-M)V_{\tau(j+1)}(r^*)+nq(0)-MV_{\tau(j+1)}(r)}, \\ &= e^{-(n-M)(V_{\tau(j+1)}(r^*)-q(0))} e^{-M(V_{\tau(j+1)}(r)-q(0))}. \end{aligned}$$

Then

$$\begin{aligned}
& \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-n(V_{\tau(j+1)}(r)-q(0))} dr \\
& \leq e^{-(n-M)V_{\tau(j+1)}(r^*)} e^{(n-M)q(0)} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
& = e^{-(n-M)\left[q(r_{\tau(j+1)}^*)-\frac{2j+2}{n}\ln(r_{\tau(j+1)}^*)\right]} e^{(n-M)q(0)} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
& = e^{-(n-M)q(r_{\tau(j+1)}^*)} r_{\tau(j+1)}^* (2j+2)\frac{n-M}{n} e^{(n-M)q(0)} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
& = e^{-(n-M)\left[\frac{q''(0)}{2}(r_{\tau(j+1)}^*)^2+O(r_{\tau(j+1)}^*)^3\right]} r_{\tau(j+1)}^* (2j+2)\frac{n-M}{n} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
& \leq e^{-\frac{(n-M)}{2}q''(0)(r_{\tau(j+1)}^*)^2} r_{\tau(j+1)}^* (2j+2)\frac{n-M}{n} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr, \\
& = e^{-c_1\frac{(n-M)}{2}r_{\tau(j+1)}^2\ln(n)^2} r_{\tau(j+1)}^* (2j+2)\frac{n-M}{n} \int_{r_{\tau(j+1)}^*}^{\infty} 2e^{-MV_{\tau(j+1)}(r)} dr := \epsilon_n(j)
\end{aligned}$$

for some $c_1 > 0$ and M as given in the other proof such that the integral converges. Since $r_{\tau} = O(\tau^{\frac{1}{2}})$, there is a constant $c_2 > 0$ such that as n goes to infinity

$$\begin{aligned}
\epsilon_n(j) &= O\left(e^{-c_1\frac{(n-M)}{2}\tau(j+1)\ln(n)^2} r_{\tau(j+1)}^* (2j+2)\frac{n-M}{n}\right), \\
&= O\left(e^{-c_2(j+1)\ln(n)^2} r_{\tau(j+1)}^* (2j+2)\frac{n-M}{n}\right), \\
&= O\left(e^{-c_2(j+1)(\ln n)^2} \left(\frac{j+1}{n}(\ln(n))^2\right)^{(j+1)\frac{n-M}{n}}\right).
\end{aligned}$$

Therefore we obtain

$$\begin{aligned}
h_j &= \int_0^{\infty} 2r^{2j+1} e^{-nq(r)} dr = \int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-nq(r)} dr + e^{-nq(0)} \epsilon_n(j), \\
&= \int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-n(q(0)+\frac{1}{2}q''(0)r^2)+O(n(r_{\tau(j+1)}^*)^3)} dr + e^{-nq(0)} \epsilon_n(j), \\
&= e^{-nq(0)} \left[\int_0^{r_{\tau(j+1)}^*} 2r^{2j+1} e^{-\frac{1}{2}q''(0)r^2n} dr \left(1 + O(n(r_{\tau(j+1)}^*)^3)\right) + \epsilon_n(j) \right], \\
&= e^{-nq(0)} \left[n^{-(j+1)} \Gamma(j+1) \left(\frac{1}{2}q''(0)\right)^{-(j+1)} \left(1 + O(n^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln n)^3)\right) + \epsilon_n(j) \right]
\end{aligned}$$

where Γ is the usual gamma function with integral representation

$$\Gamma(j+1) = \int_0^{\infty} t^j e^{-t} dt$$

and we used $t = 1/2q''(0)r^2n$.

□

5.3.2 Asymptotics of the Summation

So now we need to return to ?? and calculate each term for $\sum_{j=0}^{n-1} \ln h_j$ using the expansion in terms of [Theorem 5.5](#) and [Theorem 5.6](#) for the expression

$$\sum_{j=0}^n \ln(h_j) = \sum_{j=0}^{m_n-1} \ln(h_j) + \sum_{j=m_n}^{n-1} \ln(h_j).$$

Theorem 5.7. *We have the following asymptotics on n for the summation of $\ln(h_j)$ from 0 to $n-1$,*

$$\begin{aligned} \sum_{j=0}^{n-1} \ln(h_j) = & -n^2 I[\mu_Q] - \frac{1}{2} n \ln(n) - \frac{n}{2} (E_Q[\mu_Q] - \ln(2\pi)) - \frac{\ln(n)}{12} \\ & + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-12}(\ln n)^3) \end{aligned}$$

where $I_Q[\mu_Q]$, $F_V[D_{r_1}]$ and $E_Q[\mu_Q]$ are as in [Theorem 5.4](#).

To prove this theorem, we must give the asymptotics of both summations. This means we need both [Lemma 5.8](#), referring to the sum from powers 0 to $m_n - 1$, and [Lemma 5.9](#), referring to the sum from m_n to $n - 1$. We start by the lower power terms.

Lemma 5.8. *We have the following asymptotic for the summation of $\ln(h_j)$ from 0 to $m_n - 1$,*

$$\begin{aligned} \sum_{j=0}^{m_n-1} \ln(h_j) = & -m_n n q(0) - \frac{m_n(m_n+1)}{2} \left(\ln(n) + \ln\left(\frac{1}{2} q''(0)\right) \right) + \frac{m_n^2 \ln(m_n)}{2} - \frac{3}{4} m_n^2 \\ & + \frac{\ln(2\pi) m_n}{2} - \frac{\ln(m_n)}{12} + \zeta'(-1) + O(m_n^{-2} + n^{-\frac{1}{2}(1-5\epsilon)} (\ln(n))^3). \end{aligned}$$

where ϵ is as in the definition for m_n .

Proof. The first term in the right is straightforwardly determined, we need only to write, using [Theorem 5.6](#),

$$\begin{aligned} \sum_{j=0}^{m_n-1} \ln(h_j) = & -m_n n q(0) - \frac{m_n(m_n+1)}{2} \left(\ln(n) + \ln\left(\frac{1}{2} q^{(2)}(0)\right) \right) \\ & + \ln(G(m_n+1)) + O(n^{-\frac{1}{2}(1-5\epsilon)} (\ln(n))^3), \\ = & -m_n n q(0) - \frac{m_n(m_n+1)}{2} \left(\ln(n) + \ln\left(\frac{1}{2} q^{(2)}(0)\right) \right) + \frac{m_n^2 \ln(m_n)}{2} - \frac{3}{4} m_n^2 \\ & + \frac{\ln(2\pi) m_n}{2} - \frac{\ln(m_n)}{12} + \zeta'(-1) + O(m_n^{-2} + n^{-\frac{1}{2}(1-5\epsilon)} (\ln(n))^3) \end{aligned}$$

where G is the Barnes G -function defined recursively as

$$G(z+1) = \Gamma(z)G(z),$$

and we have used the asymptotic for the G -Barnes function

$$\ln(G(z+1)) = \frac{z^2 \ln(z)}{2} - \frac{3}{4}z^2 + \frac{\ln(2\pi)z}{2} - \frac{\ln(z)}{12} + \zeta'(-1) + \mathcal{O}\left(\frac{1}{z^2}\right).$$

□

Lemma 5.9. *We have the following asymptotic for the summation of $\ln(h_j)$ from m_n to $n-1$,*

$$\begin{aligned} \sum_{j=m_n}^{n-1} \ln(h_j) &= -n^2 I[\mu_Q] + \frac{(n-m_n)}{2} \ln \frac{2\pi}{n} + nm_n q(0) + \frac{3}{4}m_n^2 - \frac{n}{2} E_Q[\mu_Q] \\ &\quad - \frac{m_n}{2} (m_n + 1) \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{2} \left(m_n^2 - \frac{1}{6} \right) \ln \left(\frac{m_n}{n} \right) + F_V[D_{r_1}] + \mathcal{O}(n^{-12}(\ln n)^3), \end{aligned}$$

where $I_Q[\mu_Q]$, $F_V[D_{r_1}]$ and $E_Q[\mu_Q]$ are as in [Theorem 5.4](#).

The second summation is a bit more detailed, we use the previously defined asymptotics in [Theorem 5.5](#),

$$\ln(h_j) = -nV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \ln \left(\frac{8\pi r_{\tau(j)}^2}{nV_{\tau(j)}^{(2)}(r_{\tau(j)})} \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(j^{-\frac{3}{2}}(\ln(n))^\alpha)$$

and need to determine the asymptotic of three terms in order to get the full summation,

$$-n \sum_{j=m_n}^{n-1} V_{\tau(j)}(r_{\tau(j)}), \quad \sum_{j=m_n}^{n-1} \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \quad \text{and} \quad \sum_{j=m_n}^{n-1} \ln(r_{\tau(j)}).$$

This is done in [Proposition 5.3](#), [Proposition 5.4](#) and [Proposition 5.5](#).

Proposition 5.3. *The following asymptotic holds under the same context as [Lemma 5.9](#).*

$$\begin{aligned} -n \sum_{j=m_n}^{n-1} V_{\tau(j)}(r_{\tau(j)}) &= -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) + \frac{1}{6} \ln(r_1) + nm_n q(0) + \frac{3}{4}m_n^2 \\ &\quad - \frac{1}{2} \left(m_n^2 - m_n + \frac{1}{6} \right) \ln \left(\frac{4m_n}{nV_0^{(2)}(0)} \right) - \frac{m_n}{2} + \mathcal{O}\left(n^{-\frac{1}{2}(3-5\epsilon)}\right). \end{aligned}$$

Here, $I_Q[\mu_Q]$ is as in [Theorem 5.4](#), ϵ is as in the definition for m_n , and

$$2U_{\mu_Q}(0) := q(r_1) - 2\ln(r_1) - q(0).$$

Proof. We proceed as before and use the Euler-Maclaurin formula to write

$$\begin{aligned} \sum_{j=m_n}^{n-1} V_{\tau(j)}(r_{\tau(j)}) &= \int_{m_n}^n V_{\tau(t)}(r_{\tau(t)}) dt - \frac{1}{2} (V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(m_n)}(r_{\tau(m_n)})) + \\ &\quad \frac{1}{12} [\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=m_n}] + O(n^{-3}). \end{aligned}$$

Then, we are able to rewrite, setting $s = r_{\tau(j)}$,

$$\begin{aligned} \int_{m_n}^n V_{\tau(t)}(r_{\tau(t)}) dt &= \int_{m_n}^n (q(r_{\tau(t)}) - 2\tau(t) \ln(r_{\tau(t)})) dt, \\ &= \frac{1}{2} \int_{r_{\tau(m_n)}}^{r_1} (q(s) - sq'(s) \ln(s)) s V^{(2)}(s) n ds, \\ &= n \left[\frac{1}{2} \int_{r_{\tau(m_n)}}^{r_1} q(s) s V^{(2)}(s) ds - \frac{1}{2} \int_{r_{\tau(m_n)}}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds \right]. \end{aligned}$$

Using that $r_1 q'(r_1) = 2$, apply integration by parts on the second term and get

$$\begin{aligned} &\frac{1}{2} \int_{r_{\tau(m_n)}}^{r_1} s^2 q'(s) \ln(s) V^{(2)}(s) ds \\ &= \frac{1}{2} \int_{r_{\tau(m_n)}}^{r_1} s^2 q'(s) \ln(s) \frac{1}{s} (sq'(s))' ds, \\ &= \frac{1}{4} \int_{r_{\tau(m_n)}}^{r_1} ((sq'(s))^2)' \ln(s) ds, \\ &= \frac{1}{4} (sq'(s))^2 \ln(s) \Big|_{r_{\tau(m_n)}}^{r_1} - \frac{1}{4} \int_{r_{\tau(m_n)}}^{r_1} (sq'(s))^2 \frac{1}{s} ds, \\ &= \ln(r_1) - \frac{1}{4} \int_{r_{\tau(m_n)}}^{r_1} sq'(s)^2 ds - \frac{m_n^2}{n^2} \ln(r_{\tau(m_n)}), \\ &= \ln(r_1) - \frac{1}{4} (sq'(s)q(s)) \Big|_{r_{\tau(m_n)}}^{r_1} + \frac{1}{4} \int_{r_{\tau(m_n)}}^{r_1} q(s) (sq'(s))' ds - \frac{m_n^2}{n^2} \ln(r_{\tau(m_n)}), \\ &= \ln(r_1) - \frac{1}{2} q(r_1) + \frac{1}{4} \int_{r_{\tau(m_n)}}^{r_1} q(s) (sq'(s))' ds - \frac{m_n^2}{n^2} \ln(r_{\tau(m_n)}) + \frac{m_n}{2n} q(r_{\tau(m_n)}). \end{aligned}$$

Leaving us with

$$\begin{aligned} &n \left[\frac{1}{4} \int_{r_{\tau(m_n)}}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) - \frac{m_n^2}{n^2} \ln(r_{\tau(m_n)}) + \frac{m_n}{2n} q(r_{\tau(m_n)}) \right] \\ &:= nI_Q[\mu_Q] - \frac{1}{4} \int_{r_0}^{r_{\tau(m_n)}} q(s) s V^{(2)}(s) ds - \frac{m_n^2}{n} \ln(r_{\tau(m_n)}) + \frac{m_n}{2} q(r_{\tau(m_n)}). \end{aligned}$$

where we expand some terms using the following results given by [Proposition 5.2](#)

$$\ln(r_{\tau(m_n)}) = \frac{1}{2} \ln \left(\frac{4\tau(m_n)}{V_0^{(2)}(0)} \right) + O(\tau(m_n)^{\frac{1}{2}}) = \frac{1}{2} \ln \left(\frac{4\tau(m_n)}{V_0^{(2)}(0)} \right) + O(n^{-\frac{1}{2}(1-\epsilon)}), \quad (5.16)$$

$$q(r_{\tau(m_n)}) = q(0) + \frac{1}{2} q''(0) (r_{\tau(m_n)})^2 + O((\tau(m_n)^{\frac{3}{2}})) = q(0) + \tau(m_n) + O(n^{-\frac{3}{2}(1-\epsilon)}), \quad (5.17)$$

and, using integration by parts we solve the integral

$$\begin{aligned}
\frac{1}{4} \int_{r_0}^{r_{\tau(m_n)}} q(s) s V^{(2)}(s) ds &= \frac{1}{4} \int_{r_0}^{r_{\tau(m_n)}} q(s) (s q'(s))' ds, \\
&= \frac{1}{2} \tau(m_n) q(r_{\tau(m_n)}) - \frac{1}{4} \int_0^{r_{\tau(m_n)}} s (q'(s))^2 ds, \\
&= \frac{1}{2} \tau(m_n) q(r_{\tau(m_n)}) - \frac{1}{4} (q''(0))^2 \int_0^{r_{\tau(m_n)}} s^3 ds + \mathcal{O}(r_{\tau(m_n)}^5), \\
&= \frac{1}{2} \tau(m_n) q(r_{\tau(m_n)}) - \frac{1}{4} \tau(m_n)^2 + \mathcal{O}(n^{-5/2(1-\epsilon)}).
\end{aligned}$$

Therefore, for this first term, gathering the above and expanding using [Proposition 5.2](#) and (5.17),

$$\begin{aligned}
\int_{m_n}^n V_{\tau(t)}(r_{\tau(t)}) dt &= n I_Q[\mu_Q] - \frac{1}{2} \tau(m_n) q(r_{\tau(m_n)}) - \frac{1}{4} \tau(m_n)^2 + \mathcal{O}(n^{-5/2(1-\epsilon)}) \\
&\quad - \frac{m_n^2}{2n} \ln \left(\frac{4\tau(m_n)}{V_0^{(2)}(0)} \right) + \mathcal{O}(n^{-\frac{3}{2}(1-\epsilon)}) + \frac{m_n}{2} (q(0) + \tau(m_n)) + \mathcal{O}(n^{-\frac{1}{2}(1-\epsilon)}), \\
&= n I_Q[\mu_Q] - m_n q(0) - \frac{3}{4} \frac{m_n^2}{n} + \frac{m_n^2}{2n} \ln \left(\frac{4m_n}{n V_0^{(2)}(0)} \right) + \mathcal{O}(n^{-\frac{1}{2}(3-5\epsilon)}).
\end{aligned}$$

For the other terms in the Euler-Maclaurin formula we have

$$\begin{aligned}
V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(m_n)}(r_{\tau(m_n)}) &= q(r_1) - 2 \ln(r_1) - q(r_{\tau(m_n)}) + 2 \tau(m_n) \ln(r_{\tau(m_n)}), \\
&= q(r_1) - 2 \ln(r_1) - q(0) - \frac{m_n}{n} + \frac{m_n}{n} \ln \left(\frac{4m_n}{n V_0^{(2)}(0)} \right) + \mathcal{O}(n^{-3/2(1-\epsilon)}), \\
&:= 2U_{\mu_Q}(0) - \frac{m_n}{n} + \frac{m_n}{n} \ln \left(\frac{4m_n}{n V_0^{(2)}(0)} \right) + \mathcal{O}(n^{-3/2(1-\epsilon)}),
\end{aligned}$$

and,

$$\begin{aligned}
\partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=n} - \partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=m_n} &= \frac{2}{n} (\ln(r_{\tau(m_n)}) - \ln(r_1)), \\
&= \frac{1}{n} \ln \left(\frac{4m_n}{n V_0^{(2)}(0)} \right) - \frac{2}{n} \ln(r_1) + \mathcal{O}(n^{-1/2(3-\epsilon)}).
\end{aligned}$$

Finally, gathering terms, we get the lemma. □

Proposition 5.4. *We have following asymptotic*

$$\sum_{j=m_n}^{n-1} \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) = n E_Q(\mu_Q) - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - m_n \ln \left(\frac{V_0^{(2)}(0)}{4} \right) + \mathcal{O}(n^{-1/2(1-\epsilon)}),$$

where $E_Q[\mu_Q]$ is as in [Theorem 5.4](#) and ϵ is as in the definition for m_n .

Proof. Again, we start with the Euler-Maclaurin formula

$$\begin{aligned} \sum_{j=m_n}^{n-1} \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) &= \int_{m_n}^n \ln \left(\frac{V_{\tau(t)}^{(2)}(r_{\tau(t)})}{4} \right) dt - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})} \right) + \\ &\quad \frac{1}{12} \left[\partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=n} - \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=m_n} \right] + O(n^{-3}). \end{aligned}$$

We first tackle the integral on the right hand side. Changing variables $s = r_{\tau(t)}$ we have

$$\begin{aligned} \int_{m_n}^n \ln \left(\frac{V_{\tau(t)}^{(2)}(r_{\tau(t)})}{4} \right) dt &= \frac{n}{2} \int_{r_{\tau(m_n)}}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds, \\ &= \frac{n}{2} \int_{r_{\tau(m_n)}}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds, \\ &= \frac{n}{2} \left[\int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds - \int_{r_0}^{r_{\tau(m_n)}} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds \right], \\ &= \frac{n}{2} \left[2E_Q[\mu_Q] - \int_{r_0}^{r_{\tau(m_n)}} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds \right], \end{aligned}$$

where one also compute

$$\begin{aligned} \int_{r_0}^{r_{\tau(m_n)}} \ln \left(\frac{V^{(2)}(s)}{4} \right) s V^{(2)}(s) ds &= \int_{r_0}^{r_{\tau(m_n)}} \ln \left(\frac{V_0^{(2)}(0)}{4} + \frac{V_0^{(2)}(0)}{4} s + O(s^2) \right) s \left(V_0^{(2)}(0) + V_0^{(2)}(0)s + O(s^2) \right) ds, \\ &= \int_{r_0}^{r_{\tau(m_n)}} \ln \left(\frac{V_0^{(2)}(0)}{4} \right) s V_0^{(2)}(0) ds + O(r_{\tau(m_n)}^3), \\ &= \ln \left(\frac{V_0^{(2)}(0)}{4} \right) r_{\tau(m_n)}^2 V_0^{(2)}(0) + O(r_{\tau(m_n)}^3), \\ &= \frac{2m_n}{n} \ln \left(\frac{V_0^{(2)}(0)}{4} \right) + O(n^{-3(1-\epsilon)}) \end{aligned}$$

noting the definition for V and [Proposition 5.2](#). Furthermore, we know

$$\begin{aligned} \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) &= \partial_V \left(\ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \frac{1}{4} \partial_r \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right) \partial_\tau (r_{\tau(t)}), \\ &= \frac{2\partial_r V^{(2)}(r_{\tau(t)})}{nr_{\tau(t)} \left(V_{\tau(t)}^{(2)}(r_{\tau(t)}) \right)^2}, \\ &= O(n/m_n)^{-1/2}. \end{aligned}$$

Therefore, for the third term on the right hand side

$$\begin{aligned} \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=n} - \partial_t \ln \frac{1}{4} V_{\tau(t)}^{(2)}(r_{\tau(t)})|_{t=m_n} &= \frac{2\partial_r V^{(2)}(r_1)}{nr_1 \left(V_1^{(2)}(r_1) \right)^2} - O(n/m_n)^{-1/2}, \\ &= O(n/m_n)^{-1/2} = O(n^{-1/2(1-\epsilon)}). \end{aligned}$$

Then, simply expanding the series on $V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})$, we write,

$$\sum_{j=0}^{n-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = nE_Q(\mu_Q) - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - m_n \ln \left(\frac{V_0^{(2)}(0)}{4} \right) + O(n^{-1/2(1-\epsilon)}).$$

□

Proposition 5.5. *The following asymptotic holds under the same context as [Lemma 5.9](#).*

$$\sum_{j=m_n}^{n-1} \ln(r_{\tau(j)}) = n \left[-U_{\mu_Q}(0) + \frac{1}{2} \tau(m_n) \right] - \frac{1}{2} \ln(r_1) - \frac{2m_n - 1}{4} \ln \left(\frac{4\tau(m_n)}{V_0^{(2)}(0)} \right) + O(n^{-\epsilon})$$

where $U_{\mu_Q}(0)$ is as in [Proposition 5.3](#).

Proof. As usual, we start with the Euler-Maclaurin formula.

$$\sum_{j=m_n}^{n-1} \ln(r_{\tau(j)}) = \int_{m_n}^n \ln(r_{\tau(t)}) dt - \frac{1}{2} \ln \left(\frac{r_1}{r_{\tau(m_n)}} \right) + \frac{1}{12} [\partial_t \ln r_{\tau(t)}|_{t=n} - \partial_t \ln r_{\tau(t)}|_{t=m_n}]$$

where we directly note that

$$\begin{aligned} \partial_t \ln r_{\tau(t)}|_{t=m_n} &= \partial_r \ln r_{\tau(t)} \partial_r r_{\tau(t)} \partial_t \tau(t)|_{t=m_n} = \frac{2}{r_{\tau(t)}^2 V_{\tau(t)}^{(2)}(r_{\tau(t)}) n} \Big|_{t=m_n} = O(m_n^{-1}), \\ \partial_t \ln r_{\tau(t)}|_{t=n} &= \frac{2}{r_{\tau(t)}^2 V_{\tau(t)}^{(2)}(r_{\tau(t)}) n} \Big|_{t=n} = O(n^{-1}). \end{aligned}$$

Now, for the other terms we start with the first term, the integral, to get using integration by parts

$$\begin{aligned} \int_{m_n}^n \ln(r_{\tau(t)}) dt &= \frac{n}{2} \int_{r_{\tau(m_n)}}^{r_1} \ln(s) (s' q(s))' ds, \\ &= \frac{n}{2} \left[\ln(s) s q'(s) \Big|_{r_{\tau(m_n)}}^{r_1} - \int_{r_{\tau(m_n)}}^{r_1} q'(s) ds \right], \\ &= \frac{n}{2} [\ln(r_1) r_1 q'(r_1) - \ln(r_{\tau(m_n)}) r_{\tau(m_n)} q'(r_{\tau(m_n)}) - q(r_1) + q(r_{\tau(m_n)})], \\ &= \frac{n}{2} [2 \ln(r_1) - 2 \tau(m_n) \ln(r_{\tau(m_n)}) - q(r_1) + q(r_{\tau(m_n)})]. \end{aligned}$$

Ultimately, using [\(5.16\)](#) and [\(5.17\)](#),

$$\begin{aligned} \int_{m_n}^n \ln(r_{\tau(t)}) dt - \frac{1}{2} \ln \left(\frac{r_1}{r_{\tau(m_n)}} \right) + O(m_n^{-1}) &= n \left[\ln(r_1) - \frac{q(r_1) - q(0)}{2} + \frac{1}{2} \tau(m_n) \right] \\ &\quad - \frac{1}{2} \ln(r_1) - \frac{2m_n - 1}{4} \ln \left(\frac{4\tau(m_n)}{V_0^{(2)}(0)} \right) + O(n^{-\epsilon}). \end{aligned}$$

□

Finally, we turn our attention to gathering these propositions and proving the original lemma.

Proof. (Lemma 5.9) We start by stating from (5.13), with the new limits,

$$\begin{aligned} \frac{1}{n} \sum_{j=m_n}^{n-1} \mathfrak{B}(r_{\tau(j)}) &= -\frac{3}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_n)} V_{\tau(m_n)}^{(3)}(r_{\tau(m_n)})}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})} \right) \\ &\quad - \frac{1}{12} \int_{r_{\tau(m_n)}}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-2}). \end{aligned}$$

We should have by the asymptotic for h_j in Theorem 5.5, referring to $j > m_n$

$$\begin{aligned} \sum_{j=m_n}^{n-1} \ln(h_j) &= \sum_{j=m_n}^{n-1} \left(-n V_{\tau(j)}(r_{\tau(j)}) + \ln(r_{\tau(j)}) - \frac{1}{2} \ln \frac{1}{4} V_{\tau(j)}^{(2)}(r_{\tau(j)}) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) \right) \\ &\quad + \frac{(n-m_n)}{2} \ln \frac{2\pi}{n} + O(m_n^{-1/2} (\ln n)^\alpha) + O(n^{\frac{1}{2}(1-5\epsilon)}). \end{aligned}$$

By the lemmas just proved, we write, as $n \rightarrow \infty$,

$$\begin{aligned} \sum_{j=m_n}^{n-1} \ln(h_j) &= -n^2 I[\mu_Q] + \frac{(n-m_n)}{2} \ln \frac{2\pi}{n} + n m_n q(0) + \frac{3}{4} m_n^2 - \frac{n}{2} E_Q[\mu_Q] \\ &\quad - \frac{m_n}{2} (m_n + 1) \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{2} \left(m_n^2 - \frac{1}{6} \right) \ln \left(\frac{m_n}{n} \right) + \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) \\ &\quad + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_n)} V_{\tau(m_n)}^{(3)}(r_{\tau(m_n)})}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})} \right) \\ &\quad - \frac{1}{12} \int_{r_{\tau(m_n)}}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + O(n^{-12} (\ln n)^3). \end{aligned}$$

Using Proposition 5.2 we write that

$$r_{\tau(m_n)} = r_0 + O(\tau(m_n)^{\frac{1}{2}}) = O(n^{\frac{1}{2}(\epsilon-1)})$$

and

$$V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)}) = V_0^{(2)}(r_0) + O(n^{(\epsilon-1)}).$$

Then, we rewrite

$$\begin{aligned} \sum_{j=m_n}^{n-1} \ln(h_j) &= -n^2 I[\mu_Q] + \frac{(n-m_n)}{2} \ln \frac{2\pi}{n} + n m_n q(0) + \frac{3}{4} m_n^2 - \frac{n}{2} E_Q[\mu_Q] \\ &\quad - \frac{m_n}{2} (m_n + 1) \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{2} \left(m_n^2 - \frac{1}{6} \right) \ln \left(\frac{m_n}{n} \right) + F_V[D_{r_1}] + O(n^{-12} (\ln n)^3). \end{aligned}$$

where

$$F_V[D_{r_1}] := \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - \frac{1}{32} \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12} \int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

□

And here finally we return to the full summation.

Proof. (Theorem 5.7) Gathering Lemma 5.8 and Lemma 5.9, we get

$$\begin{aligned} \sum_{j=0}^{n-1} \ln(h_j) &= -n^2 I[\mu_Q] - \frac{1}{2} n \ln(n) - \frac{n}{2} (E_Q[\mu_Q] - \ln(2\pi)) - \frac{\ln(n)}{12} \\ &\quad + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-12}(\ln n)^3). \end{aligned}$$

Incredibly, all the terms depending on m_n cancels out.

□

We finally prove Theorem 5.4.

Proof. (Theorem 5.4) We restate the previous result

$$\begin{aligned} \sum_{j=0}^{n-1} \ln(h_j) &= -n^2 I_Q[\mu_Q] - \frac{1}{2} n \ln n - \frac{n}{2} (E_Q[\mu_Q] - \ln(2\pi)) - \frac{\ln n}{12} \\ &\quad + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-1/12}(\ln n)^3). \end{aligned}$$

We get the asymptotic expansion for the Free energy noting that all terms involving m_n vanish and using that

$$\ln(n!) = n \ln(n) - n + \frac{1}{2} \ln(n) + \frac{1}{2} \ln(2\pi) + O(n^{-1})$$

that is,

$$\begin{aligned} \ln \mathcal{Z}_n^Q &= \ln n! + \sum_{j=0}^{n-1} \ln h_j \\ &= -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + n \left(\frac{\ln(2\pi)}{2} - 1 - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{5}{12} \ln n \frac{\ln(2\pi)}{2} \\ &\quad + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-1/12}(\ln n)^3). \end{aligned}$$

□

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